

The background of the slide is a photograph of a historic building, likely a university building, with a prominent clock tower and multiple windows. The building is partially obscured by large, leafy trees on either side. A semi-transparent white rectangular box is overlaid on the left side of the image, containing the title and author information.

Supercapacitors

A. Balducci

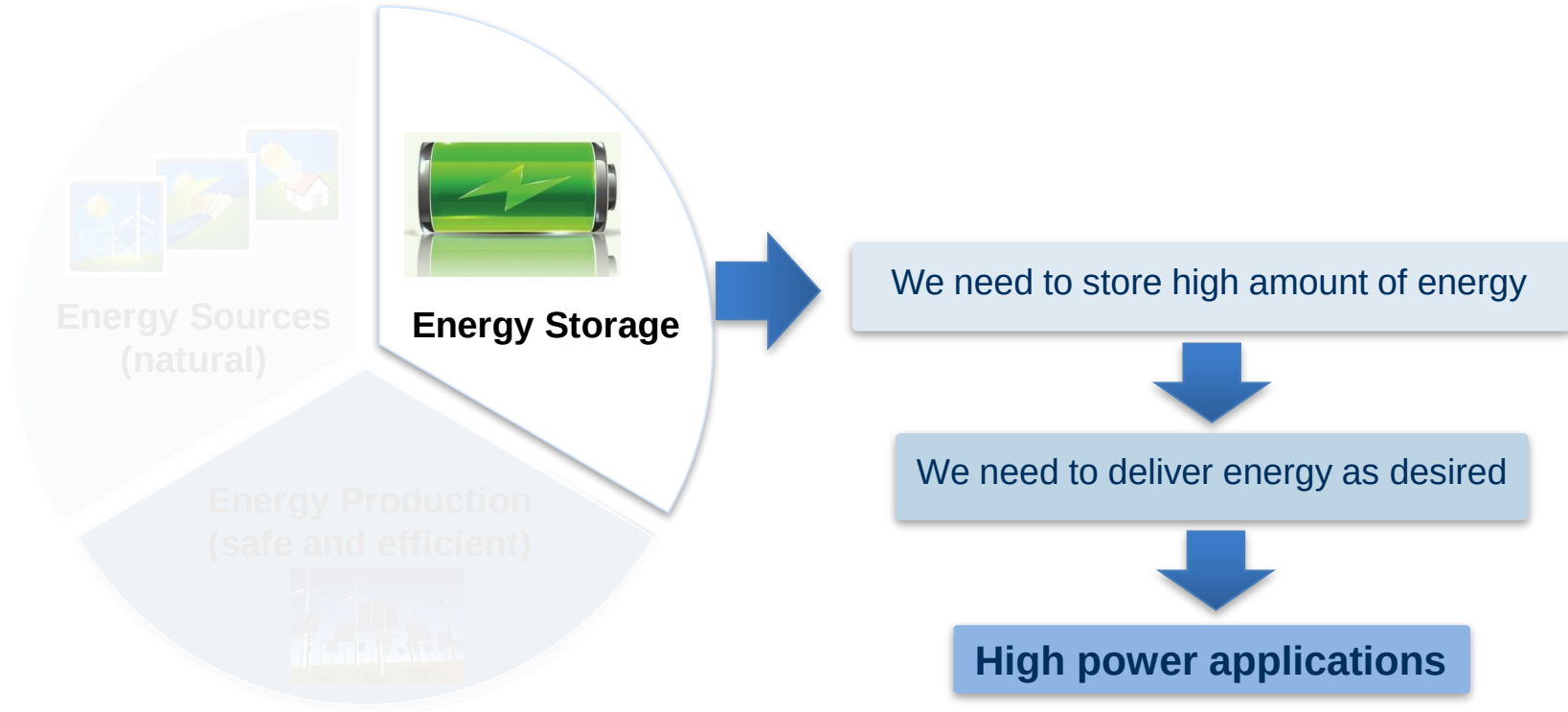
Outline

- High power applications: supercapacitors vs. batteries
- Supercapacitors
 - Components and materials
 - Techniques
- EDLC
 - Carbons
 - Electrolytes
- Pseudocapacitors
- Hybrid devices

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Energy delivering



Energy on demand

“Energy on demand”: An effective delivery is essential for an effective storage

In our daily life we are facing many different situations, and each of them requires a different **amount of energy** and, also, a different **delivery time**

There are situations in which we need:

Fast delivery/uptake: minutes or second

Repeated delivery: tens of thousands or million of times



High power applications

High power applications



Fast deliver of energy and high cycle life

High power applications

High power for emergency situations

Back-up systems in hospitals



Emergency doors in aircrafts



Fast delivery of energy, but long life is not a requirement

High power applications in transportation

“Stop and go” function



“Stop and go” in heavy vehicles



Electric vehicles



Peak power



Fast delivery of energy and high cycle life

High power applications

High power is required in public transportation



Trains



Buses

They can be charged at the station

Fast delivery / uptake of energy and high cycle life

High power applications in stationary applications

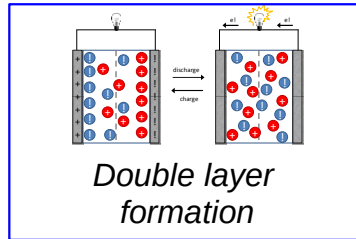


Fast delivery might be necessary to optimize the use of alternative energy sources and stabilize (smart) grid

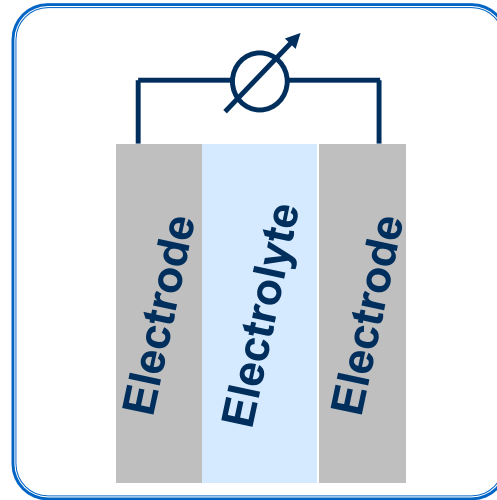
High power and electrochemical devices

Electrochemical energy storage devices

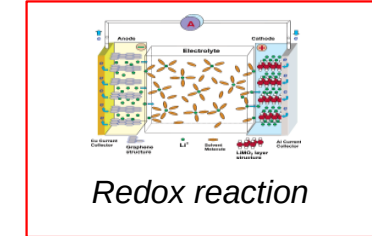
Physical storage



Supercapacitors

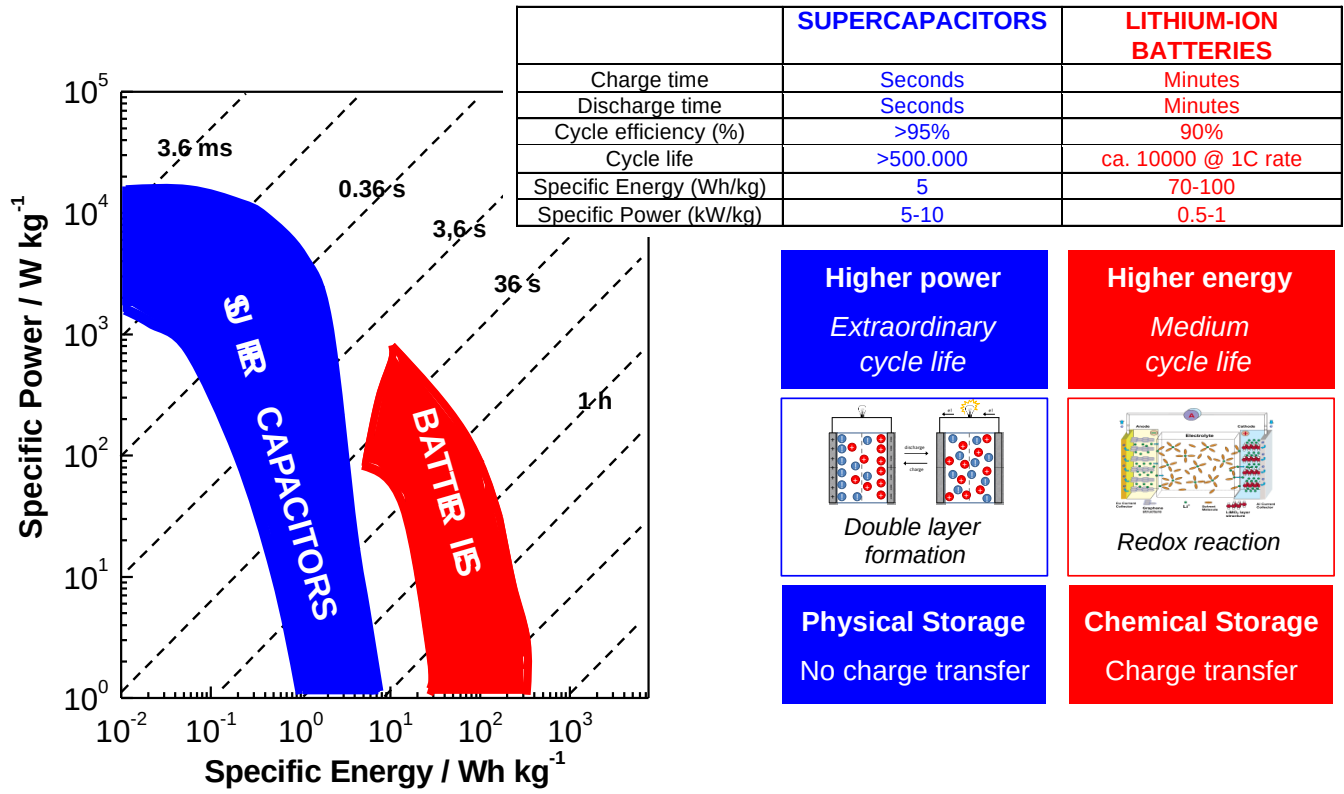


Chemical storage



Lithium-ion batteries

EDLCs vs. LIBs



EDLCs applications

Electrochemical Capacitors: Challenges and Opportunities for Real-World Applications

by John R. Miller and Andrew F. Burke

Interface 2008

Journal of Power Sources 326 (2016) 726–735



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Journal of Power Sources

journal homepage: www.elsevier.com/locate/jpowsour



Engineering electrochemical capacitor applications



John R. Miller

JME, Inc. and Case Western Reserve University, 23500 Mercantile Road, Suite K, Beachwood, OH 44122, USA

H I G H L I G H T S

- Electrochemical capacitors are used today in a broad range of applications.
- Approaches presented for the design of capacitor storage systems.
- No single commercial product offers superior performance in all applications.

EDLCs vs. LIBs

City bus with “stop and go” driving: thousand start/stop during a day



High cycle life
Need to dissipate the heat generation
due to charge-discharge losses



Supercap: 3000F, 2,7 V
High power battery: 12Ah

FIG. 2. Hybrid city transit bus that uses an electrochemical capacitor energy storage system. Typical charge/discharge periods may be forty seconds in this application with continuous 16-hours-per-day operation (>1000 charge/discharge cycles each day). (Photo courtesy of ISE Corporation.)

EDLCs vs. LIBs

Supercap: 3000F, 2,7 V

High power battery: 12Ah

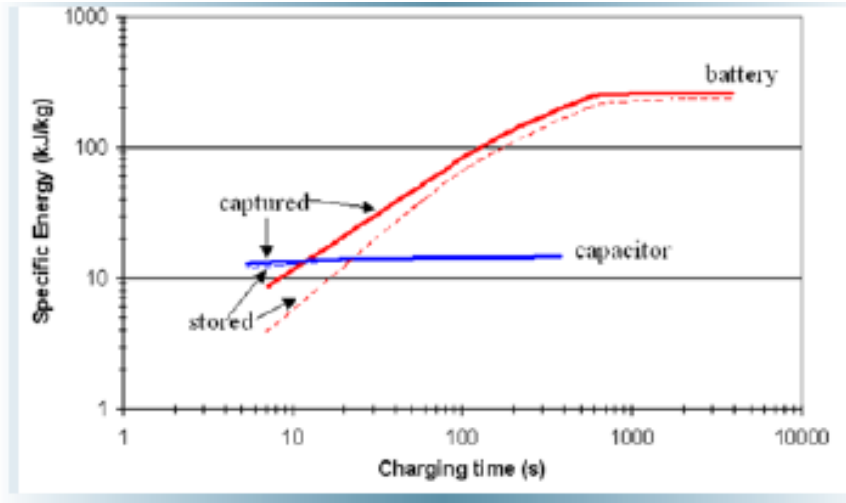
Test performed by measuring the amount of energy captured by each device for various constant-current charge times

The stored regenerative energy was the amount of energy that could be discharge after the charge

Charging:
2,7 V for supercap
2,8 V for battery

Initial state
Supercapacitor 1,35 V
Battery 10% SOC

EDLCs vs. LIBs



$t > 600$ s (10 minutes)

Battery has a constant ca. 15 times higher specific energy than the capacitors

$t = 100$ s battery capture ca. 5 times more energy than capacitor

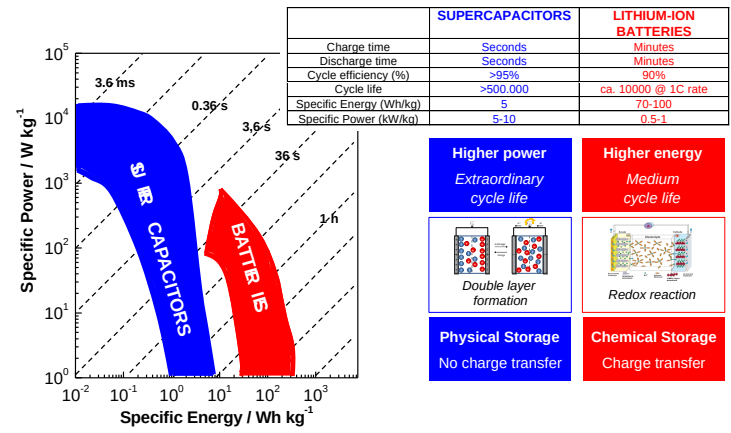
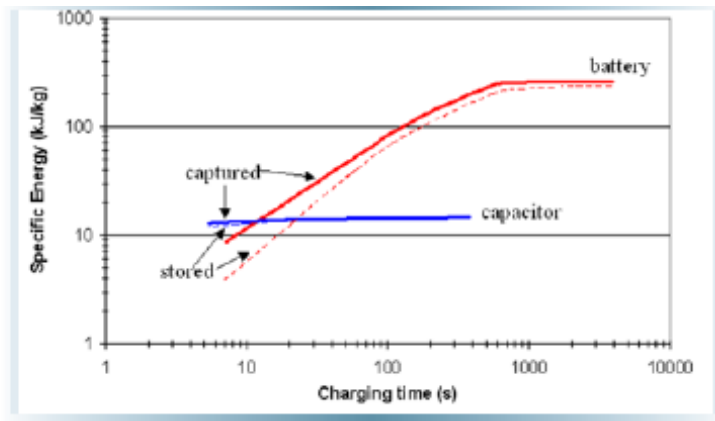
$t = 10$ s battery and supercap capture the same specific energy.

BUT, supercap can discharge ca. 95% of this amount, while the battery only ca. 50% of the amount.

The effective energy density of the capacitor is twice that of the battery (that will need larger cooling system to remove the dissipated energy)

$T < 10$ s supercap better performance (specific energy remain flat well below 10 s)

EDLCs vs. LIBs



Supercaps more appropriate for high power application

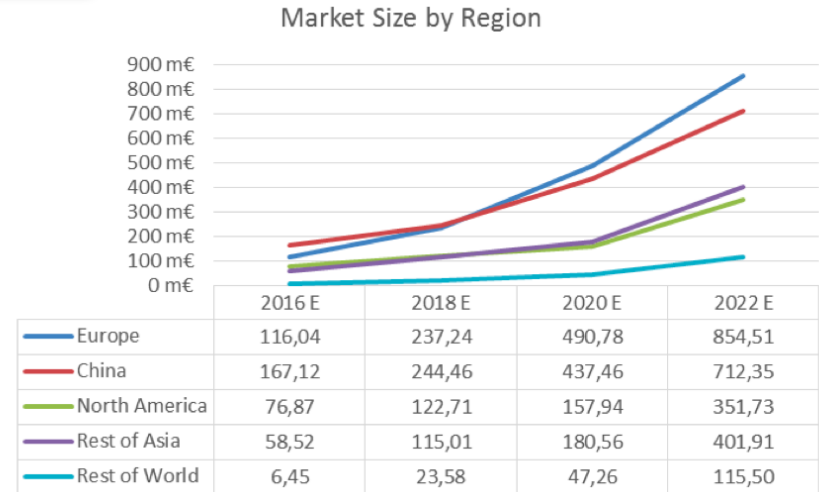
Battery not well suited for high-rate cyclic operation
Only a relatively small fraction of the battery's energy can be utilized

High power applications

There is a strong need of high power systems

High power systems need to display:

- High power
- Relatively high energy
- Short charge-discharge time
- High cycle life
- Reliability
- Large applicability
- High safety

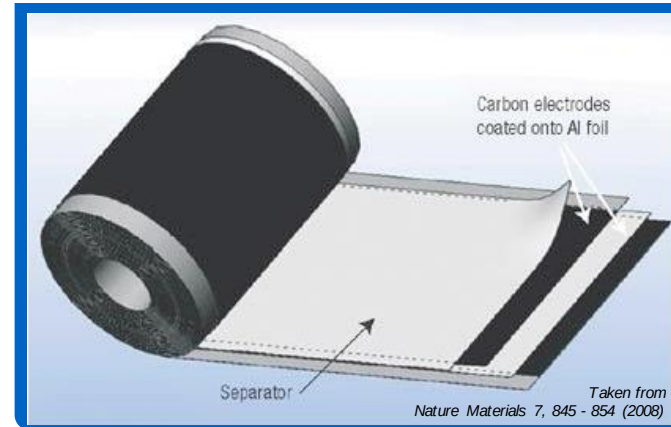


Source:IDTechEx

***Global market supercapacitor in 2016
was ca. 450 million USD***

Supercapacitor producers

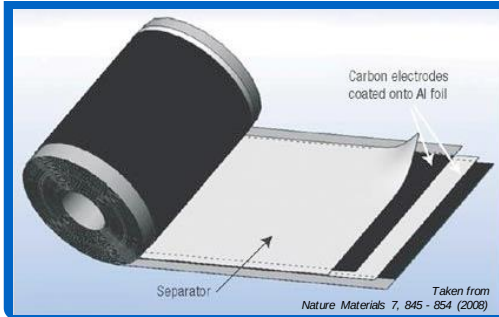
- Maxwell (USA)
- Ioxus (USA)
- JSR Micro (Japan)
- Nippon Chemi-Con Corporation(Japan)
- Nesscap (Korea)
- Cap-XX (Australia)
- Yunasko (Russia)
- Skeleton (Estonia-Germany)



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Supercapacitor components



CELL COMPONENTS

Casing

ELECTRODE COMPONENTS

Active material

Conducting agent

Binder

Current collectors

ACTIVE MATERIALS

INACTIVE MATERIALS

ELECTROLYTE COMPONENTS

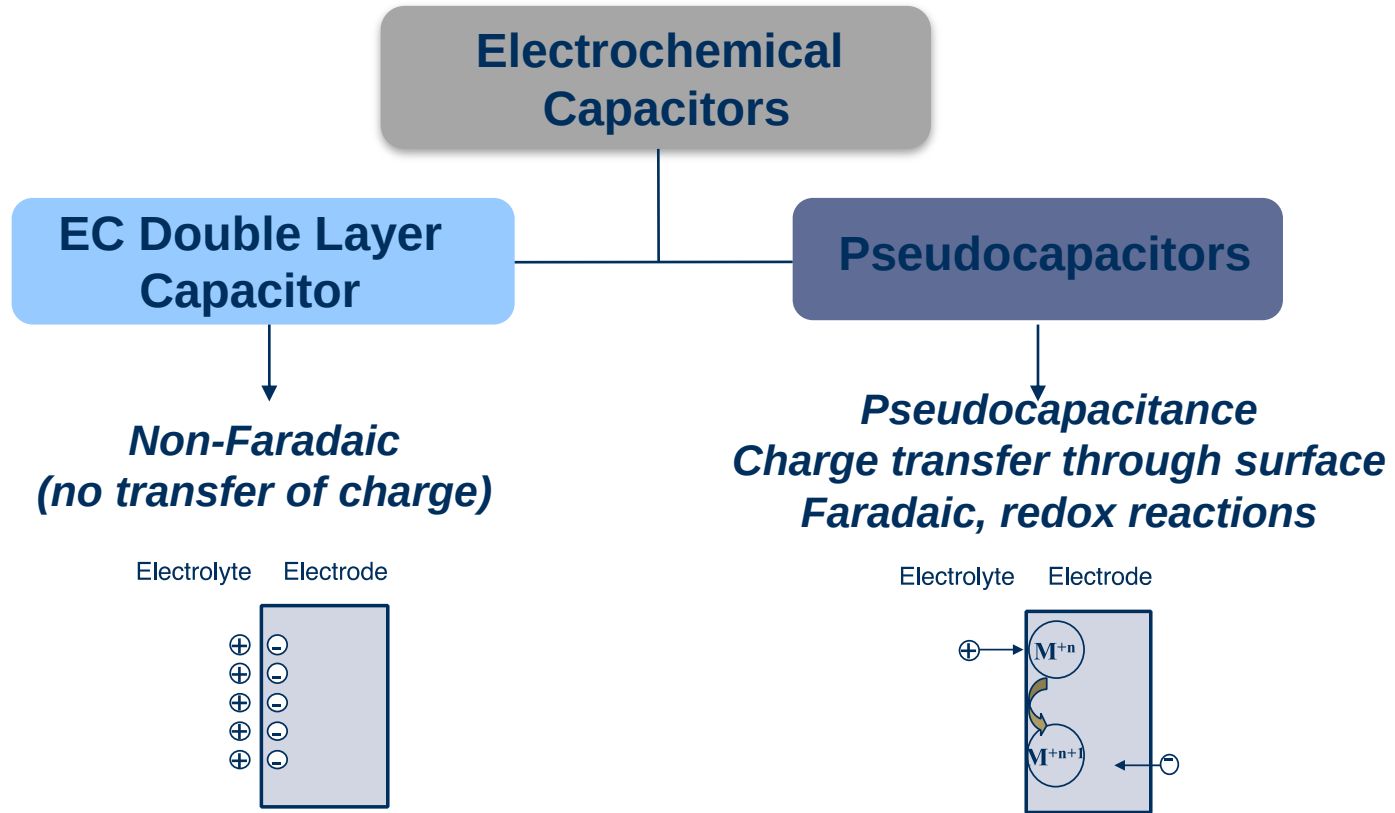
Solvent

Conducting salt

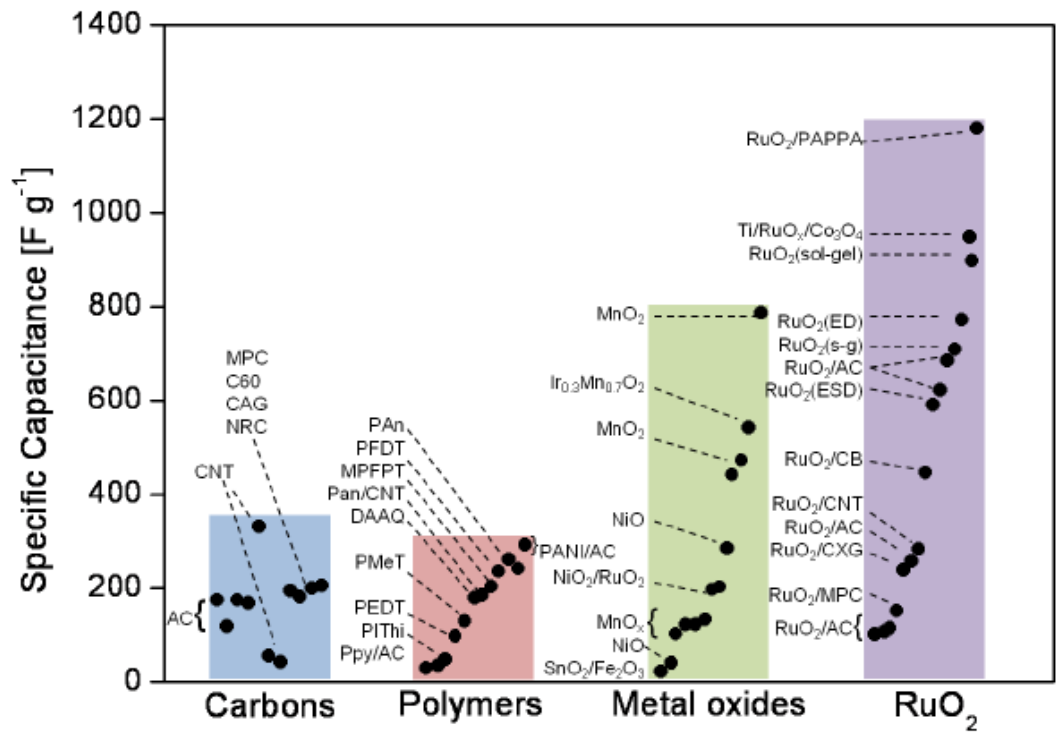
Supercapacitor classification

- Active material
- Type of electrolyte
- “Degree of hybridization”

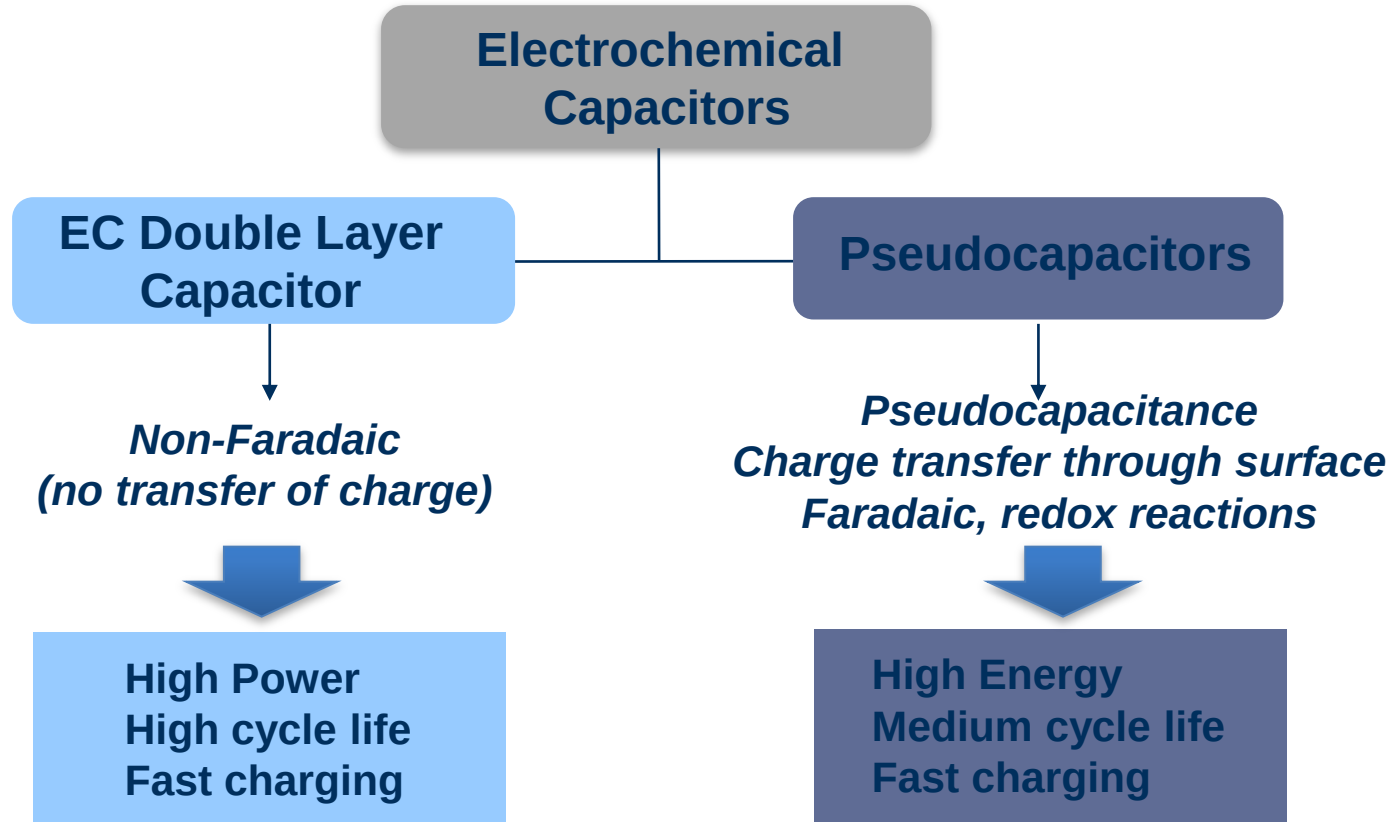
Supercapacitors



Active materials in supercapacitors



Supercapacitors



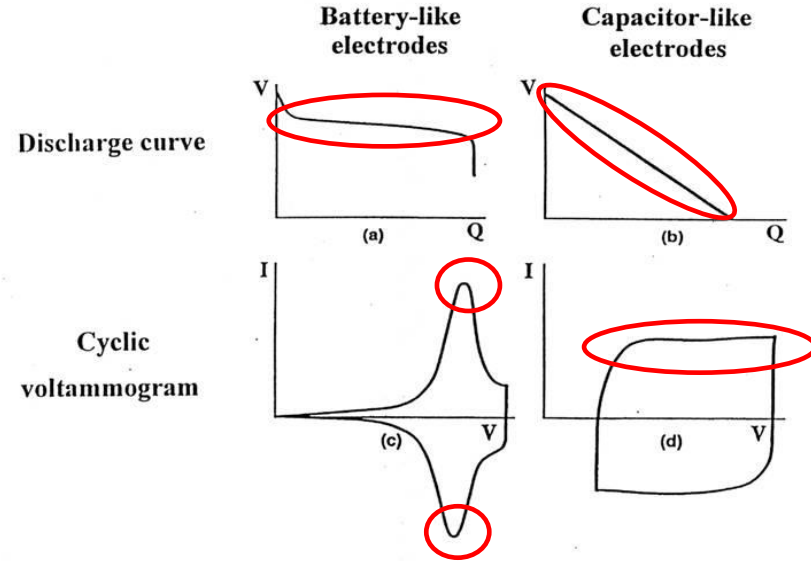
Capacitor-like and battery-like electrodes

Capacitor-like electrodes

The charge discharge process is limited only by the electrochemical stability of the electrolyte

Battery-like electrodes

The charge discharge process take place ONLY at well defined potential



Energy and power

$$E_{\max} = \frac{1}{2} CV^2$$

$$P_{\max} = \frac{V^2}{4R}$$

C = Capacitance

V = Maximum supercap voltage

R = Equivalent Series Resistance (ERS)

- Maximum supercapacitor voltage ($\Delta V_{\max, Sc}$)

$$\Delta V_{\max, Sc} = V_{\max, +} + |V_{\max, -}|$$

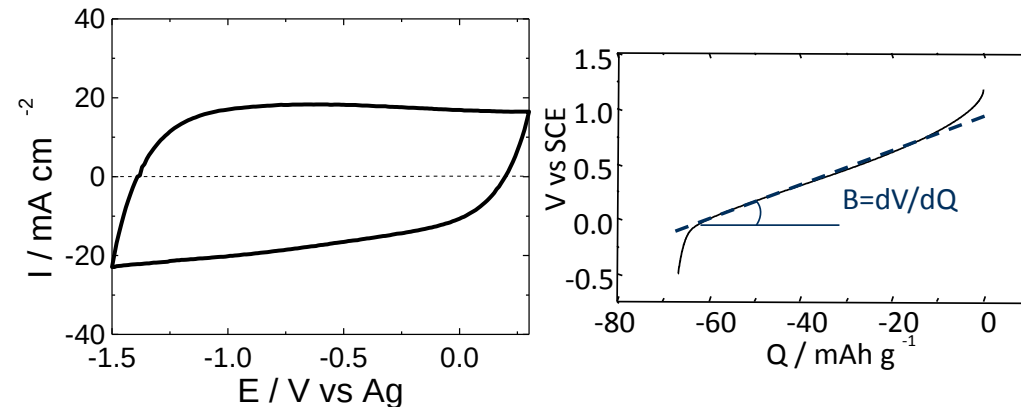
- Supercapacitor specific capacitance (C_{Sc})

$$C_{Sc} = \left(\frac{1}{C_+ w_+} + \frac{1}{C_- w_-} \right)^{-1} / w_{Sc}$$

- Equivalent Series Resistance (ESR)

$$ESR = R_{\text{ionic}} + R_{\text{electronic}}$$

Performance metrics (from CV)



$i \text{ (A/cm}^2\text{)}$ = current density
 $s \text{ (V/s)}$ = scan rate
 $m \text{ (g/cm}^2\text{)}$ = electrode mass

SPECIFIC CAPACITANCE

a) $C \text{ (F/g)} = i/(s \times m)$

b) $C \text{ (F/g)} = 3.6 \times dQ/dV = 3.6/B$

SPECIFIC CAPACITY

$Q \text{ (mAh/g)} = \int i dV / (s \times m \times 3.6)$

COULOMBIC EFFICIENCY

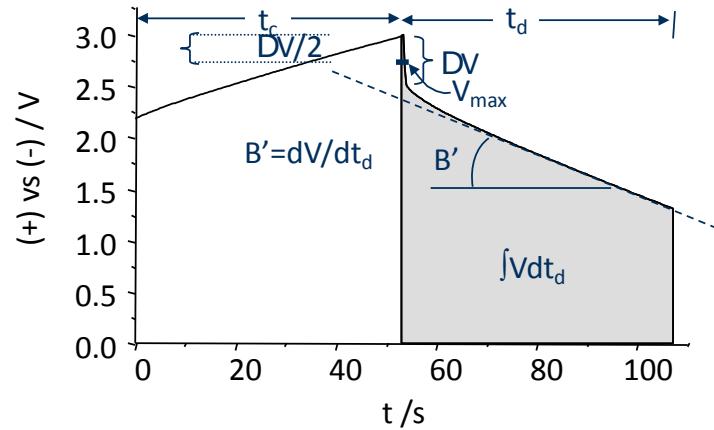
charge-discharge process

$h\% = (Q_{\text{discharge}} / Q_{\text{charge}}) \times 100$

CYCLING STABILITY

$\% \text{ capacity fade} = [1 - (Q_{1000\text{th cycle}} / Q_{10\text{th cycle}})] \times 100$

Performance metrics (from CC)



t_c (s) = time of charge

t_d (s) = time of discharge

w_{act} (g/cm²) = total active mass

w_{comp} (g/cm²) = total composite mass

i (A/cm²) = current density

V_{max} (V) = maximum cell voltage

ESR (Wcm²) = $DV/(2 \times i)$

DEVICE CAPACITANCE

$$C \text{ (F/cm}^2\text{)} = i \text{ (dt}_d\text{/dV)} = i/B'$$

MAXIMUM SPECIFIC POWER

$$P_{max} \text{ (W/kg)} = V_{max}^2 / (4 * ESR * w_{comp})$$

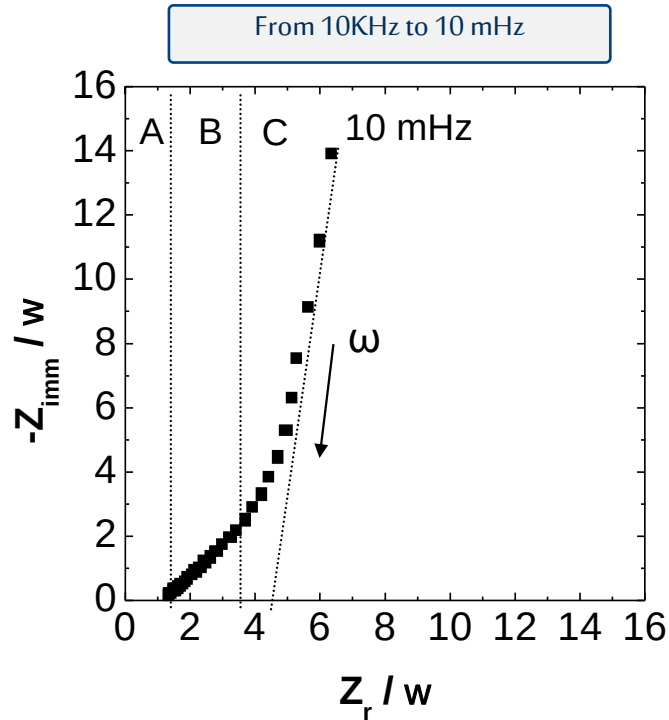
COULOMBIC EFFICIENCY

of the charge-discharge process $h\% = t_d/t_c \times 100$

CYCLING STABILITY

$$\% \text{ capacity fade} = [1 - (Q_{1000th \text{ cycle}} / Q_{10th \text{ cycle}})] \times 100$$

Performance metrics (from impedance spectroscopy)



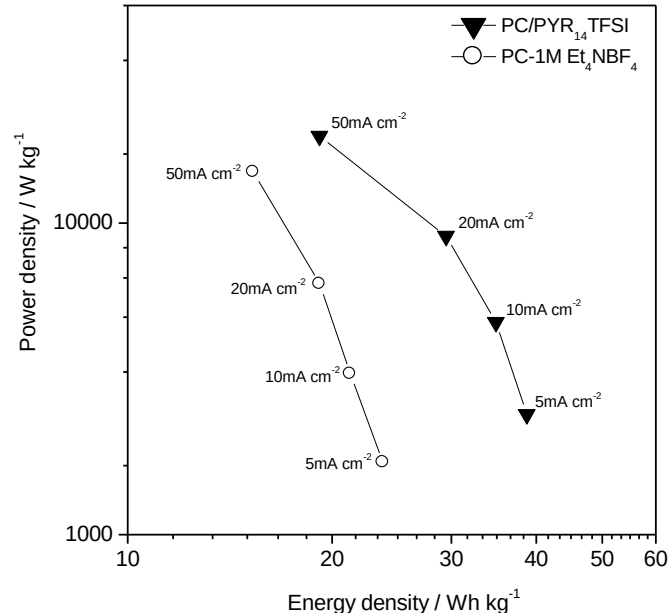
A (high frequency)= pure resistivity behaviour
Uncompensated resistance (bulk) $Z_{imm}=0$

B (intermediate frequency)= the signal “penetrate” more deeply into the porous structure of the electrodes.
The capacitive behaviour appears in the diagram

C (low frequency)= the capacity behaviour dominate and it is visible in the linear part of the diagram.
Limit capacitance can be calculated using:

$$C = \frac{1}{2\pi n Z_{imm}}$$

Energy and power



AVERAGE SPECIFIC ENERGY

$$E_{av} \text{ (Wh/kg)} = i \times \int V dt_S / (w_{comp} \times 3.6)$$

AVERAGE SPECIFIC POWER

$$P_{av} \text{ (W/kg)} = E_{av} \times 3600 / t_S$$

t_S (s) = discharge time

w_{comp} (g/cm²) = total mass composite electrode materials per cm²

i (A/cm²) = current density



A Guideline for Reporting Performance Metrics with Electrochemical Capacitors: From Electrode Materials to Full Devices

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^dInstitut des Matériaux Jean Rouxel (IMN), Université de Nantes, CNRS, 44322 Nantes Cedex 3, France

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Over the past decade, interest in electrochemical capacitors as an energy-storage technology has increased enormously, spurring the development and evaluation of a large number of new materials and device configurations. This perspective article aims to propose guidelines by which new materials and devices should be evaluated, and how resulting data should be reported with respect to critical metrics such as capacitance, energy and power.

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MATERIALS SCIENCE

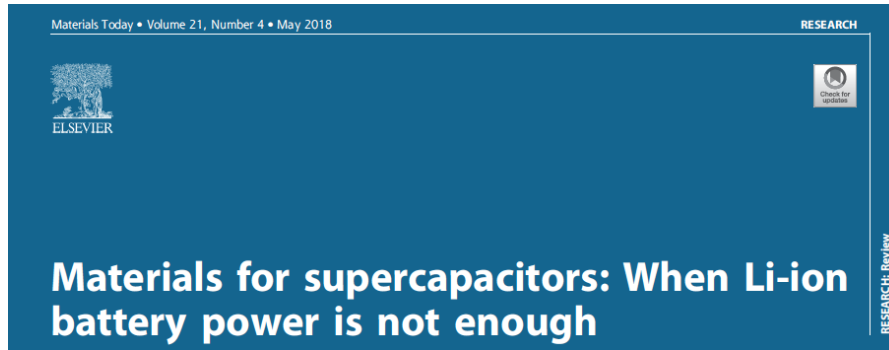
True Performance Metrics in Electrochemical Energy Storage

Y. Gogotsi¹ and P. Simon²

Exceptional performance claims for electrodes used in batteries and electrochemical capacitors often fail to hold up when all device components are included.

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 - Carbons
 - Electrolytes
- Pseudocapacitors
- Hybrid devices



Z. Lin^{1,2}, E. Goikolea^{3,4}, A. Balducci^{5,6}, K. Naoi^{7,8}, P.L. Taberna^{1,2}, M. Salanne^{2,9},
G. Yushin¹⁰, P. Simon^{1,2,*}

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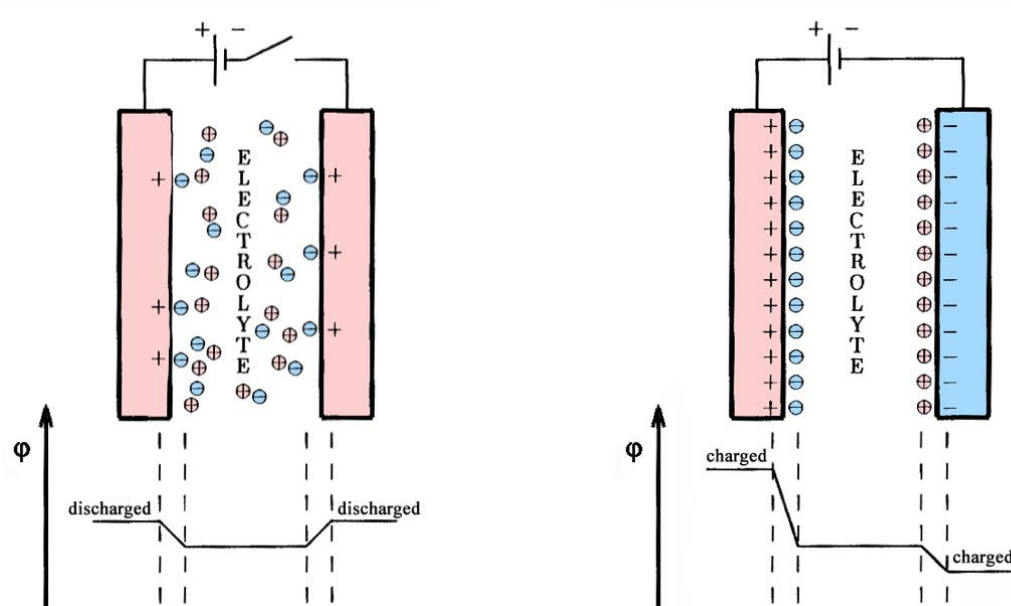
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Carbons and Electrolytes for Advanced Supercapacitors

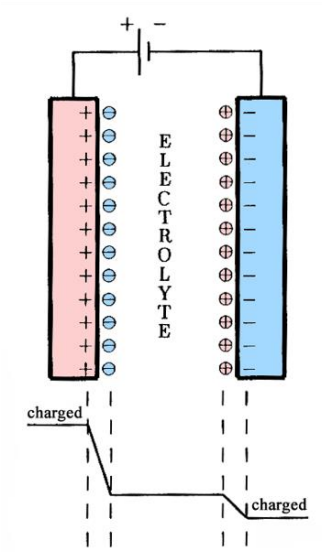
François Béguin, Volker Presser, Andrea Balducci, and Elzbieta Frackowiak*

EDLCs

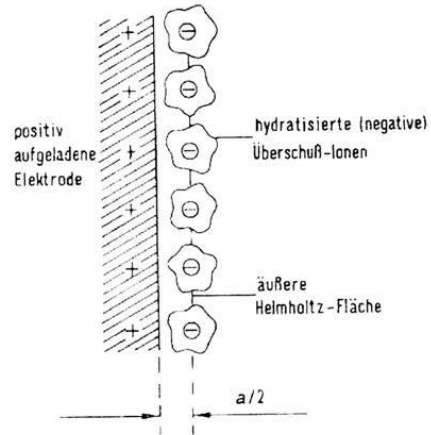
Electrochemical double layer capacitors (EDLCs) store the electric energy in an electrochemical double layer formed at the solid/electrolyte interface



Electrical double layer



Helmholtz model



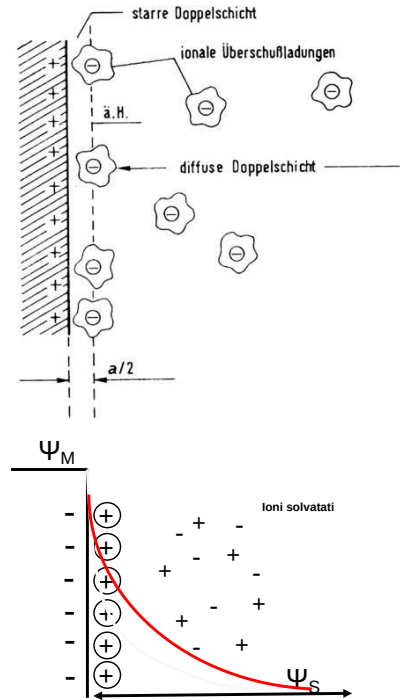
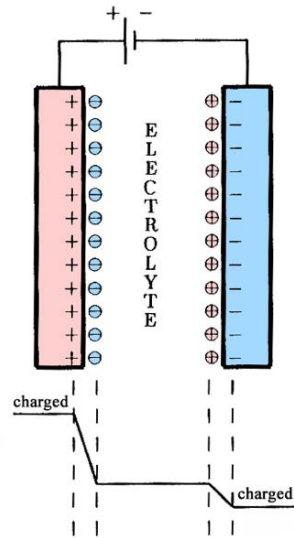
Bidimensional

Describe the excess of charge of solvated ions at the interface metal/electrolyte.

Distance between solvated ions and metal is comprise between 0.2 and 0.5 nm

After, the double layer is called external layer

Electrical double layer



Gouy-Chapman model

Tridimensional

Introduce a factor related with the thermal fluctuation that consider the movement of the ions

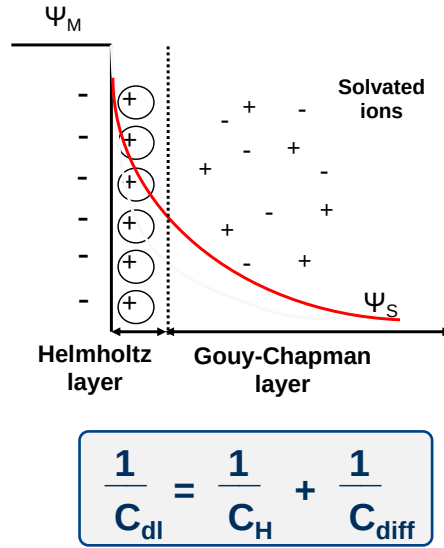
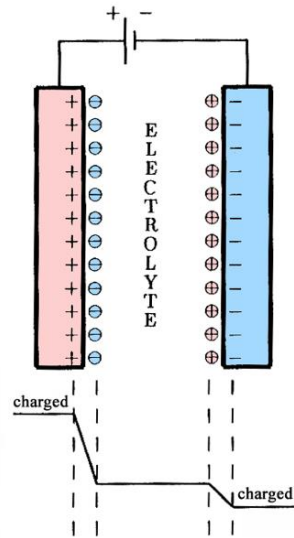
It is called diffuse layer

Describe the diffusion of the excess of charge near the electrode inside the electrolytic solution.

The ions are considered as punctual charges.

The potential associated with the diffuse layer vary exponentially with the distance from the interface

Electrical double layer



Stern model

Tridimensional

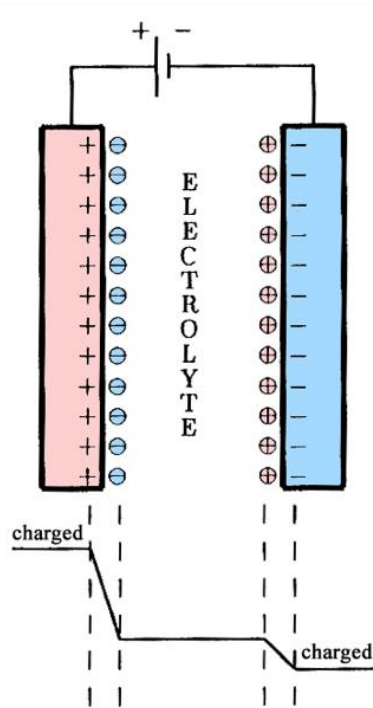
Describe the simultaneously presence of a diffuse and compact layer.

Solvated ions close to the electrode are absorbed following a Langmuir isotherm

The external region is described by a Gouy-Chapman layer

The double layer capacitance is limited by the smaller capacitance contribution and change with the potential electrode and the ions concentrations

EDLCs: storage process



Electrochemical capacitors store the electric energy in an electrochemical double layer formed at the solid/electrolyte interface



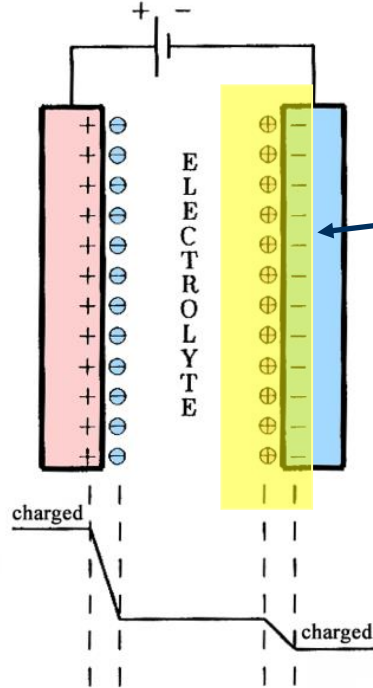
The charge-discharge process is limited only by the electrochemical stability of the electrolyte



When the electrolyte is not longer electrochemically stable faradic reactions will take place

EDLCs: storage process

Electrostatic charge process



Positive and negative IONIC charges within the electrolyte accumulate at the surface of the solid electrode and compensate for the electronic charge at the electrode surface

The thickness of the DL depend on the concentration of the electrolytes and on the size of the ions
(for concentrated electrolytes is in the order of 5-10Å)

The double layer **CAPACITANCE** can be estimated as:

$$C = \epsilon_r \epsilon_0 A / d$$

C = capacitance

ϵ_r = electrolyte dielectric constant

ϵ_0 = dielectric constant of the vacuum

A = electrode surface area

d = effective thickness of the double layer (charge separation distance)

EDLCs: storage process

$$C = \epsilon_r \epsilon_0 A / d$$

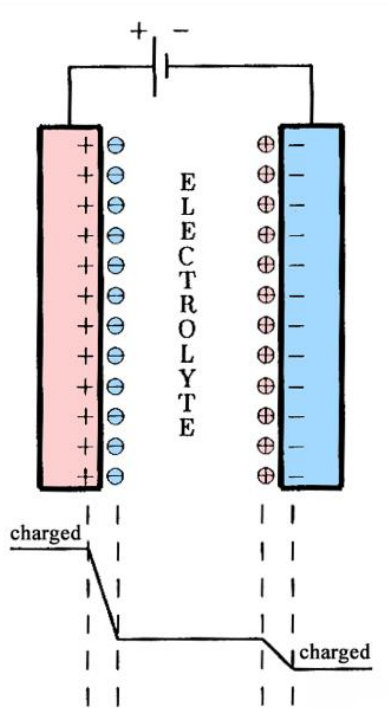
The double layer capacitance is about 10-20 $\mu\text{F}/\text{cm}^2$ for a smooth electrode in concentrated electrolytic solutions

The corresponding electric field in the electrochemical double layer is very high and assumes values up to 10^6 V/cm easily

Compared to conventional capacitors the energy density stored in the electrochemical double layer display is high (Ragone plot)

In order to achieve a higher capacitance the electrode surface area is additionally increased using porous electrodes with a extremely large internal effective surface

HIGH SURFACE AREA MATERIALS



Carbons in EDLC

The attraction of carbons as a supercapacitors electrode materials arise from a unique combination of chemical and physical properties:

- High conductivity
 - High surface-area range (ca. 1 to > 2000m²/g)
 - Good corrosion resistance
 - High temperature stability
 - Controlled pore structure
 - Processability and compatibility in composite materials
 - Relatively low cost
- *Activated carbon*
 - Graphene
 - Carbon nanotubes
 - CDC
 -

Activated carbon

Carbons can be converted into a form that has very high surface area

Activation is the process employed to increase surface area and porosity from a carbonised organic precursor (“CHAR”).

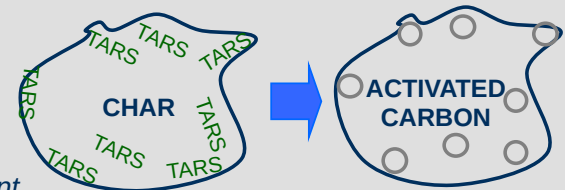
The resulting materials are referred to as **ACTIVATED CARBONS**

CHAR have low porosity and their structure consists of elementary crystallites with a large number of interstices.

The interstices tend to be filled with disorganized carbon residues (“**TARS**”) that block the pore entrance

Activation opens the pores and can create additional porosity

- Carbon precursor
- Activation condition
 - Temperature
 - Time
 - Gaseous environment



Control over the porosity, pore-size distribution, nature of internal surface

Activated carbon

Thermal activation (physical activation)

Modification of carbon char by controlled gasification

Usually between 700 – 1100°C in presence of oxidising gases

During gasification the oxidizing atmosphere greatly increases the pores volume and surface area of the materials through a controlled carbon “burn-off” and the elimination of volatile pyrolysis products

- Burn-off level influence the quality of the activated carbon
- It is controlled by the temperature and duration of activation

Chemical activation

Usually between 400 – 700°C.

Involves the dehydrating action of agent like KOH, H₃PO₄, ZnCl

Post activation washing of the carbon is usually required to remove residual reactant and inorganic residues

Activated carbon

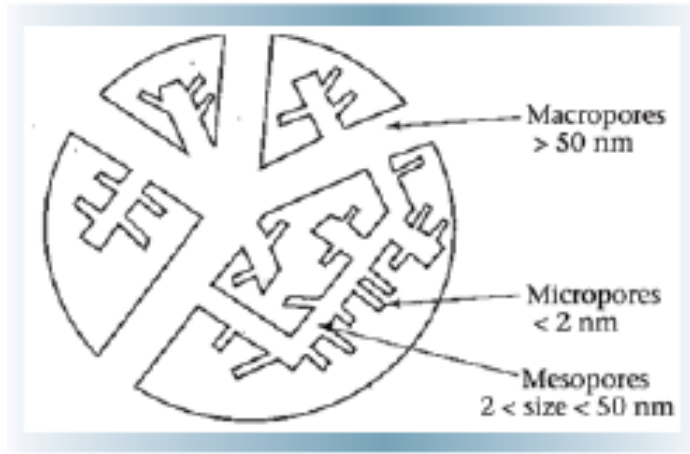


Fig. 1. Schematic diagram of the pore size network of an activated carbon grain.

Macropores > 50nm

Mesopores $2 < \text{size} < 50 \text{ nm}$

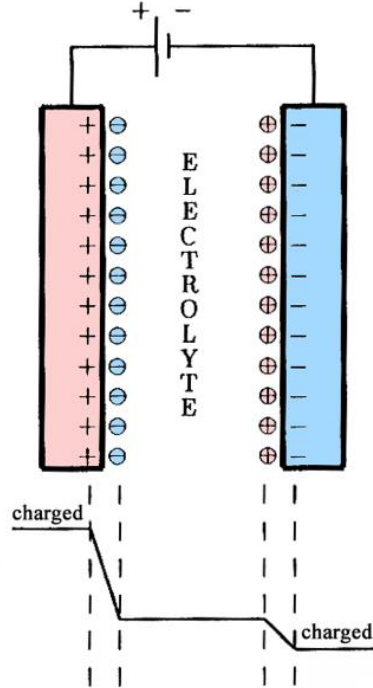
Micropores > 2nm

Micropores give the higher contribution to the surface area

Mesopores contribute to the surface area and allows improvement accessibility

Macropores negligible contribution to the surface, but they act as transport avenue

Activated carbon



$$C = \epsilon_r \epsilon_0 A/d$$



Is it the capacitance (F/g) proportional to its available surface area (m²/g)?



NO

Major factor that contribute to capacitance:

- Electrode surface area
- Morphology
- Surface chemistry
- Measurement condition

Carbon and capacitance

Intrinsic resistivity (intra-particle) of a carbon material is dependent on its chemical and structural morphology

In electrodes electrical resistance depend:

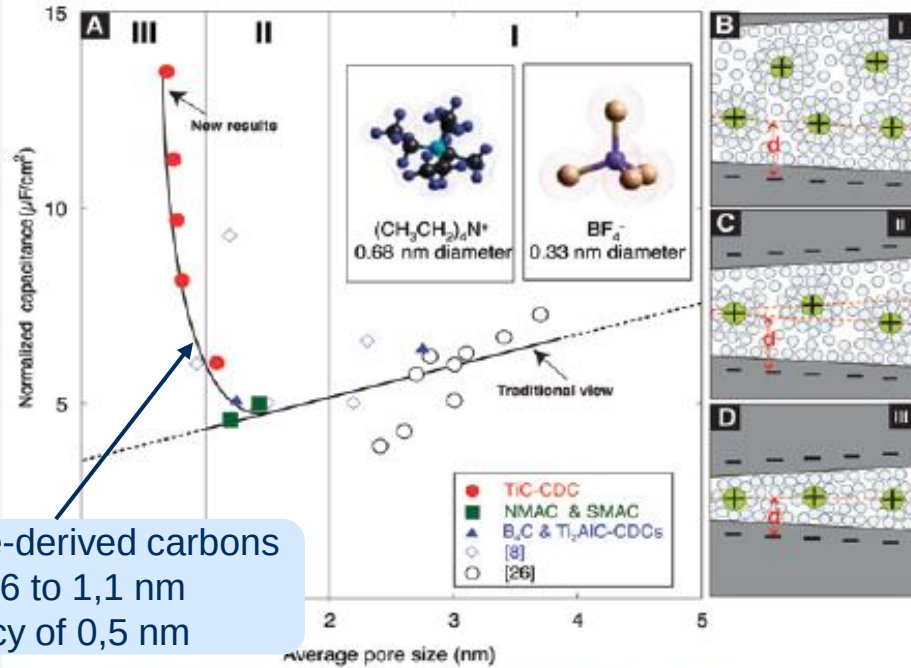
- Intra-particle resistance
- Inter-particle resistance (contact)



Resistance associated with:

- Current-carrying path
- Across carbon/carbon interface
- Across carbon/metallic current collector
- Through the metallic current collector

Carbon and capacitance



Traditional view

*Distortion of the ion solvation shell or even desolvation
Close approach of the ion to the carbon surface*

FIG. 2. Normalized capacitance vs. average pore size for TiC-CDC and other carbons from literature tested in the same electrolytes.^{23,16} (From Ref. 15.)

Carbon and capacitance

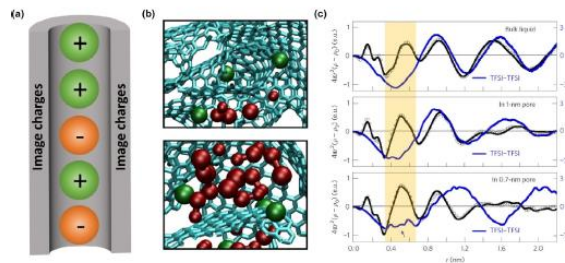


FIGURE 4
Formation of a superionic state in electrified nanopores: from theory to experiments. (a) Mean field-theory have first predicted that the presence of image charges at the surface of the nanoporous carbon exponentially screen out the electrostatic interactions of ions, leading to the packing of ions of the same sign; (b) molecular dynamics simulations of a supercapacitor formed with a CDC electrode and a BM-PP6 ionic liquid have displayed the formation of such structures (green: PF₆⁻ anions, red: BMIM⁺ cations, turquoise sticks: carbon-carbon bonds). Reproduced with permission from Ref. [128]; (c) Scattering experiments have finally confirmed the existence of anion-anion correlations in the nearest-neighbor region when they are adsorbed in 0.7-nm pores. Reproduced with permission from Ref. [129].

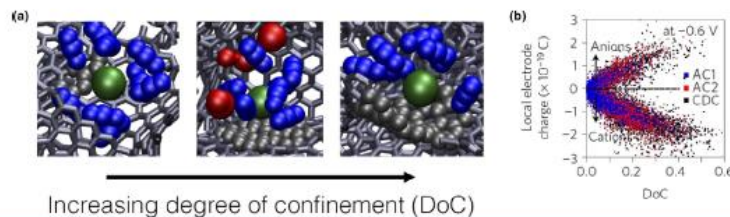


FIGURE 5
Impact of the degree of confinement of the ions on the local surface charge. (a) Molecular dynamics simulations have evidenced the existence of various adsorption sites for the ions inside nanoporous carbons, for which a degree of confinement (DoC) can be defined. Here PF₆⁻ anions (green spheres) adsorbed on sites with increasing DoC are shown from left to right (blue: ACN molecules, red: 1-butyl-3-methylimidazolium cations, gray spheres and sticks: carbon electrode). Reproduced with permission from Ref. [130]; (b) In situ small-angle X-ray scattering experiments have confirmed the correlation between the magnitude of the local electrode charge with the DoC of the adsorbed ions for a CsCl aqueous electrolyte adsorbed in various nanoporous carbons (AC1, AC2 and CDC). Reproduced with permission from Ref. [131].

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COMMUNICATION

Capacitance in carbon pores of 0.7 to 15 nm: a regular pattern†

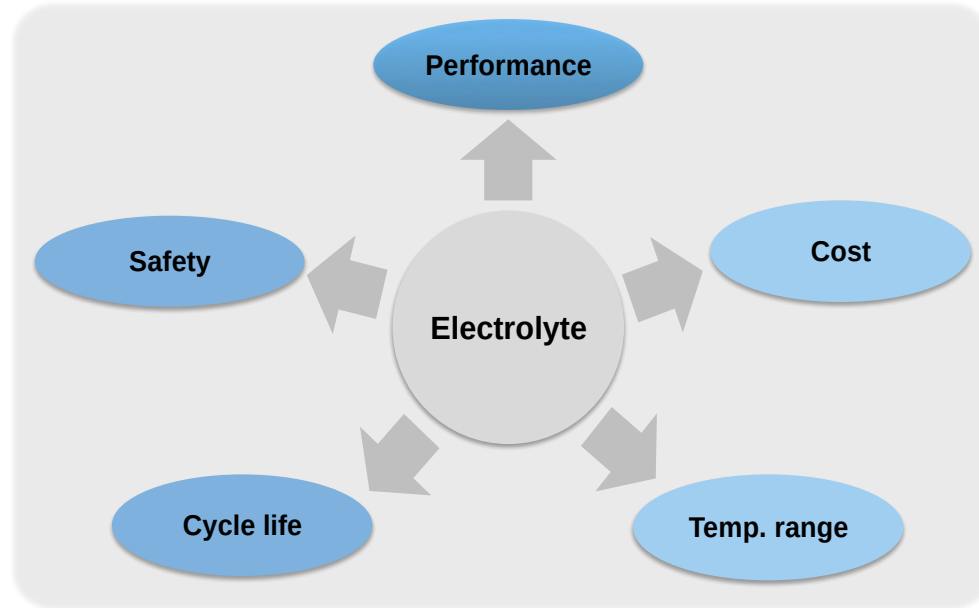
Teresa A. Centeno,^a Olha Sereda^b and Fritz Stoeckli^{a,b}

Received 14th March 2011, Accepted 20th May 2011

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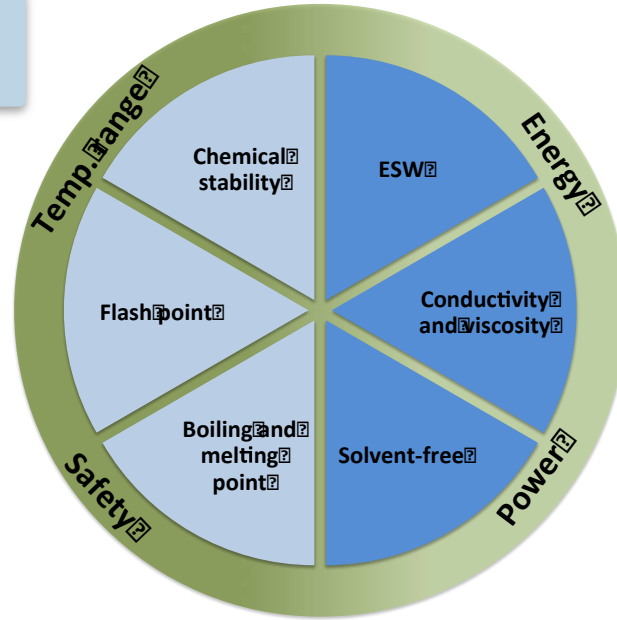
Electrolytes

Electrolytes is a key component of electrochemical storage devices



Ideal electrolyte for EDLCs

Electrolyte is
an active material



Ideal electrolyte

- Large electrochemical stability
- Solvent-free
- Low melting and high boiling points
- High thermal and chemical stabilities
- High flash point
- *Cheap (Cost)*

Electrolytes for EDLCs: classes

- Aqueous electrolytes
- Organic electrolytes
- Ionic liquids

Density: aqueous < organic < ionic liquids

Conductivity: aqueous > organic > ionic liquids

Cell Voltage: aqueous < organic < ionic liquids

For the same activated carbon

C in aqueous electrolyte (170 F g⁻¹)

C in organic electrolyte (130 F g⁻¹)

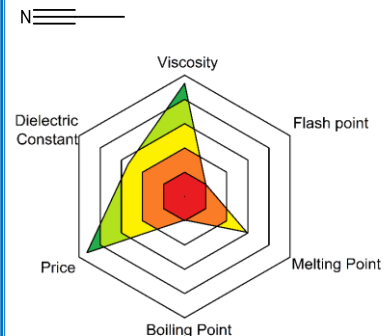
C in ionic liquid (100 F g⁻¹)

Properties of various electrolytes.			
Electrolyte	Density (gm/cm ³)	Resistivity (Ohm-cm)	Cell Voltage
KOH	1.29	1.9	1.0
Sulfuric acid	1.2	1.35	1.0
Propylene carbonate	1.2	52	2.5-3.0
acetonitrile	.78	18	2.5-3.0
Ionic liquid	1.3-1.5	125 (25°C) 28 (100°C)	4.0 3.25

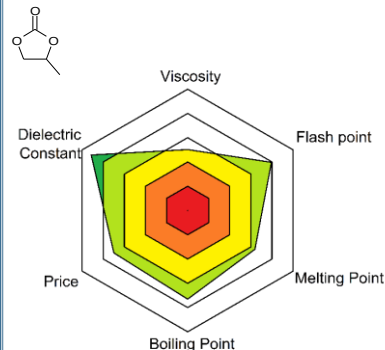
Electrolytes in EDLCs: state of the art

Solvents

Acetonitrile (ACN)

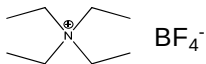


Propylene Carbonate (PC)



Conducting salt

Ammonium salt



State of the art

- High conductivity
- Good operative temperature range of use
- **Cell voltage: 2.7 – 2.8 V**

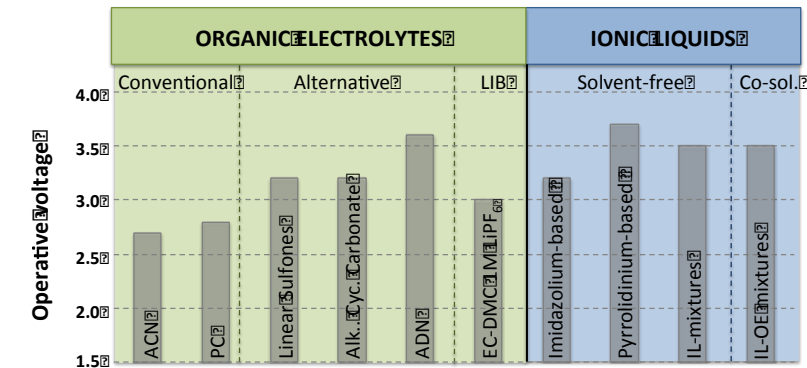
$$E_{\max} = \frac{1}{2} CV^2 \quad P_{\max} = \frac{V^2}{4R}$$

Current electrolytes cannot be used for the realization of high voltage EDLC



Alternative electrolytes are needed

Alternative electrolytes

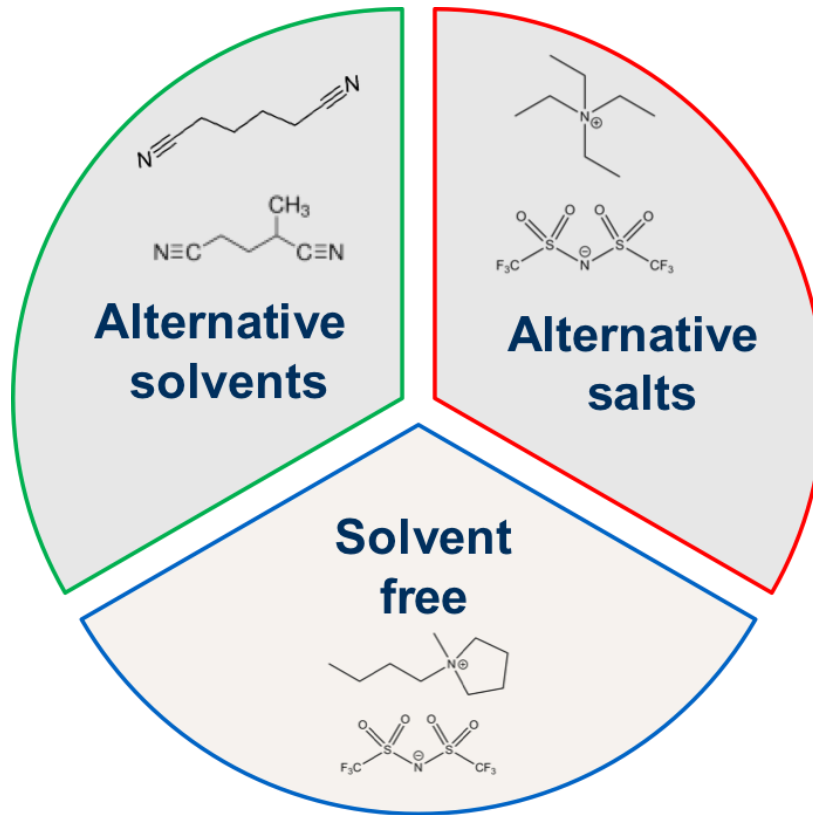


Conventional electrolytes for EDLCs		
Solvents		
- Acetonitrile (ACN) - Propylene Carbonate (PC)		
Salts		
Cation		
Tetra-ethylammonium (Et₄N⁺)		
Anion		
Tetra fluoroborate (BF₄⁻)		
Alternative electrolytes for EDLCs		
Solvents		
Nitriles		
- Glutaronitrile (GTN) - Adiponitrile (ADN) - Butyronitrile (BTN) 2-Methylglutaronitrile (2-MGN)		
Carbonate		Ester and Cyano ester
- Butylene Carbonate (BC) - Ethylene Carbonate (EC)		3-cyanopropionic acid methyl ester (CPAME) - Methyl acetate (MA) - Methyl formate (MF)
Sulfonate		
Ethyl isopropyl sulfone (EIPS)		
Salts		
Cation		
- Pyrrolidinium (Pyr) - Imidazolium (Im) - Azepanoum (Azp) - Sulfonium (Me₃S) - Cyclic sulfonium (CS)		
Anion		
- Bis(trifluoromethansulfonyl)imide (TFSI) - Bis(fluorosulfonyl)imide (FSI) - Dicyanamide (DCA) - Tricyanomethanide (C(CN)₃⁻) - Hexafluorophosphate (PF₆⁻)		

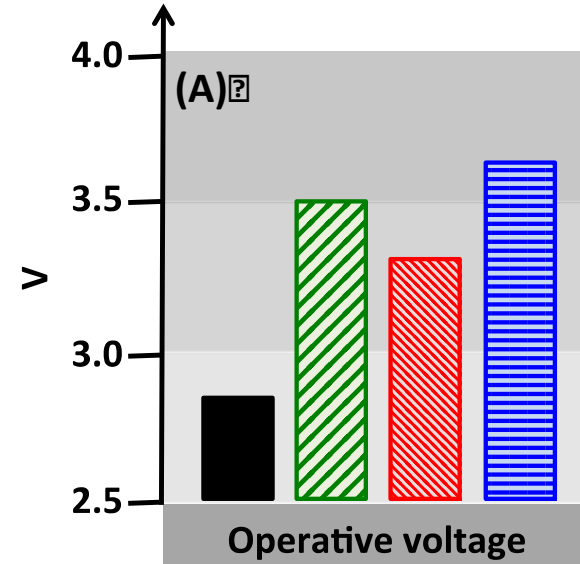
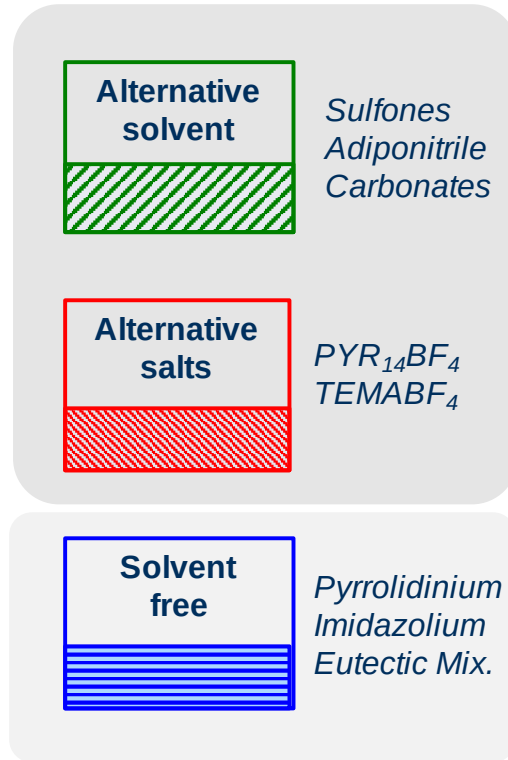
FIGURE 2

Overview of the solvents and salts utilized in conventional and alternative electrolytes for EDLCs.

Electrolyte components

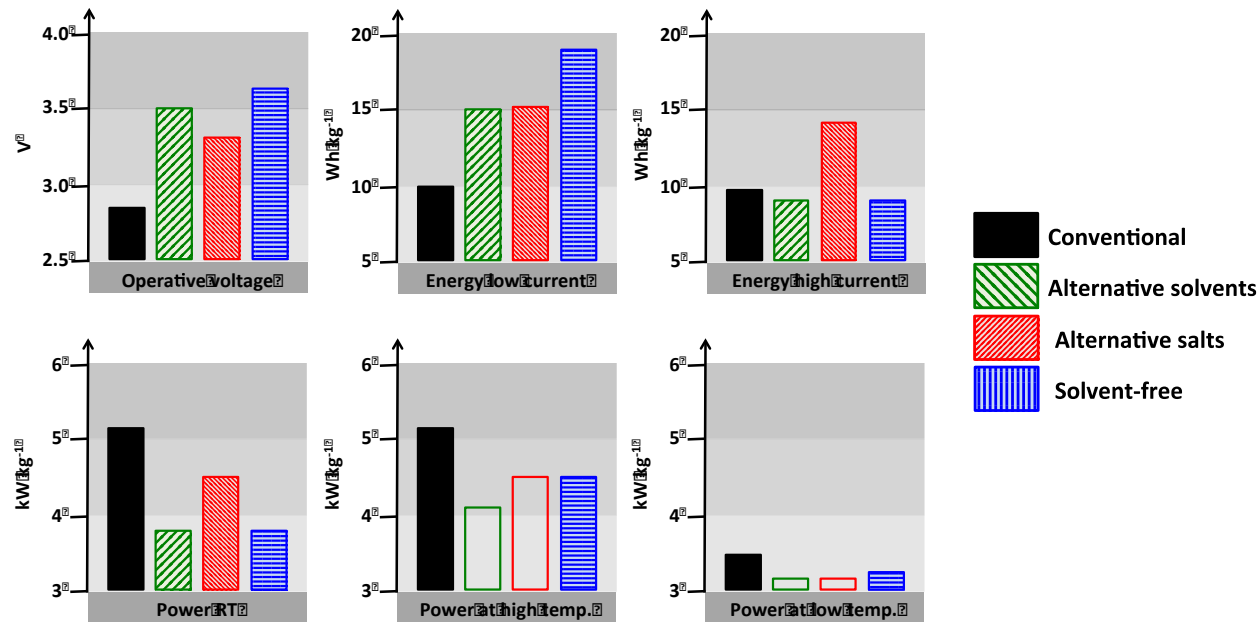


Operative voltage in alternative electrolytes



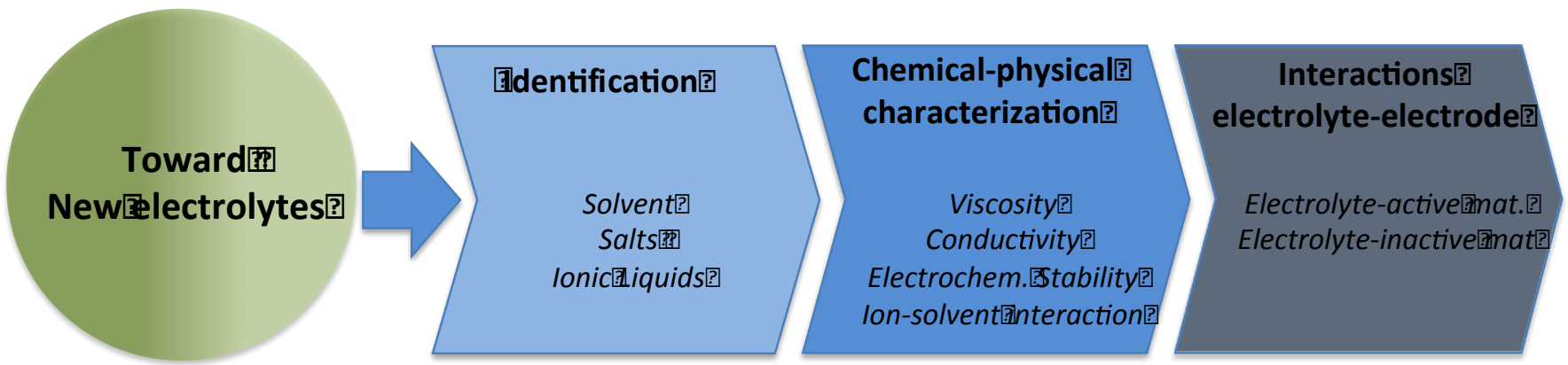
The use of alternative electrolytes might lead to important increase of the operative voltage of EDLCs

Energy and power in alternative electrolytes

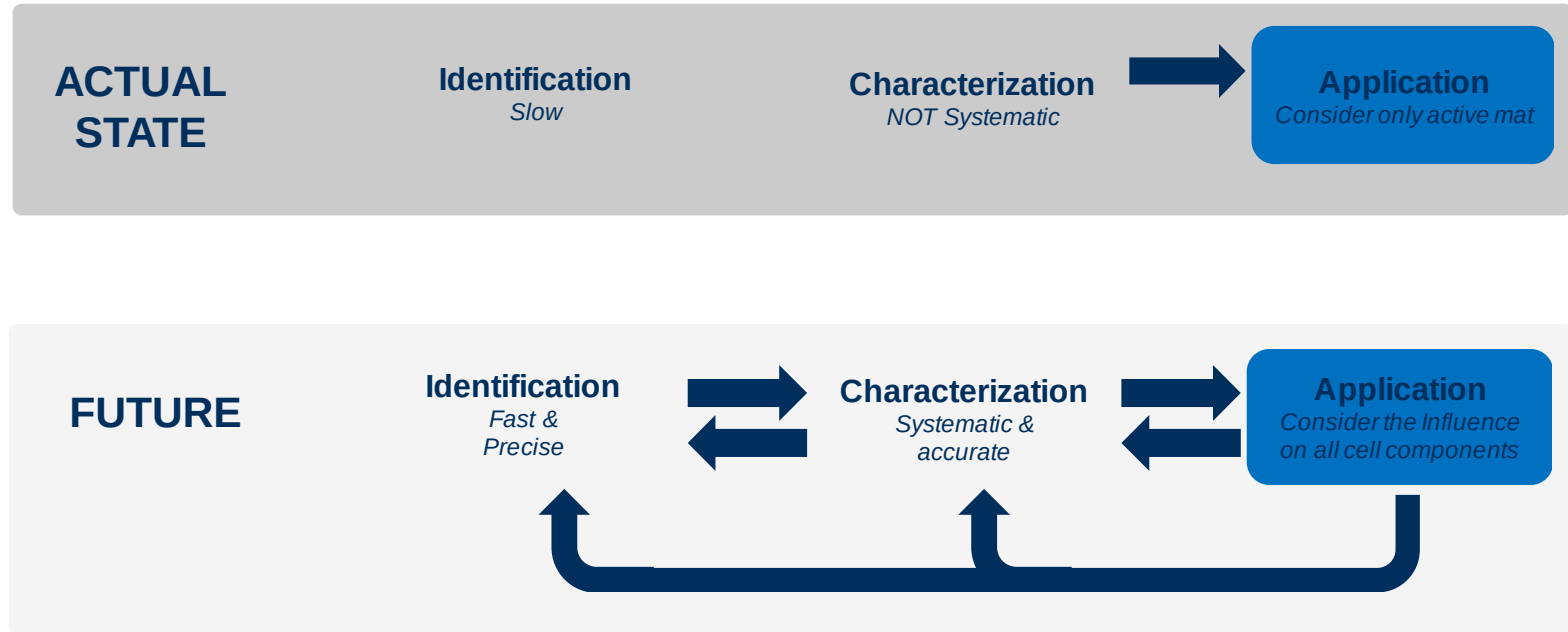


The overall performance of EDLCs containing alternative electrolytes is not higher than those of systems containing conventional electrolytes

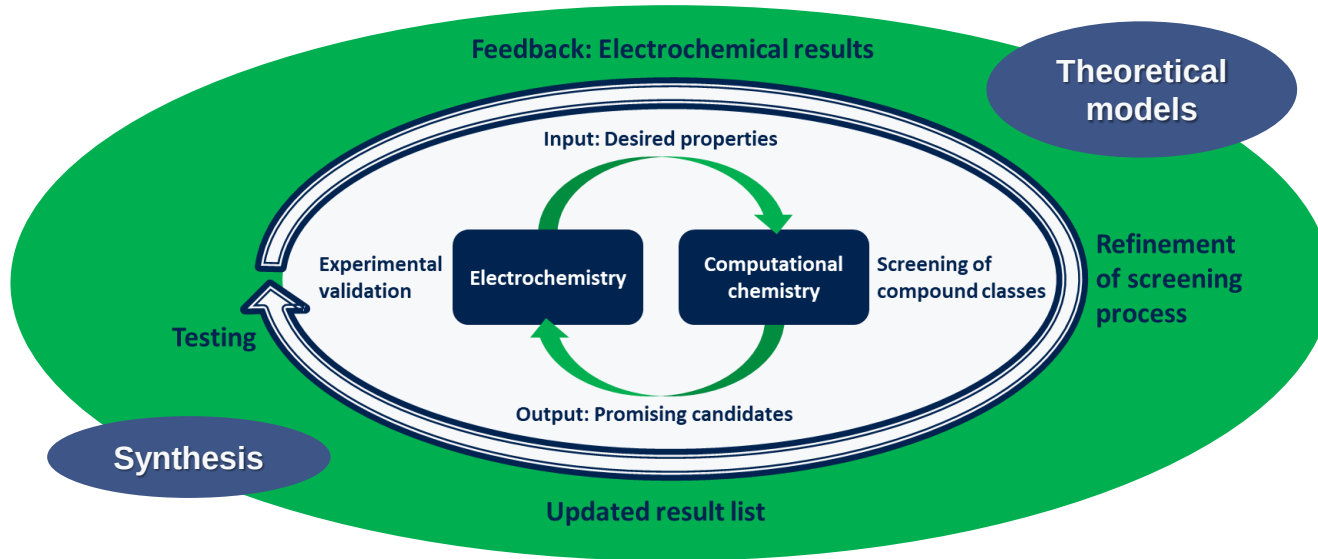
Development of novel electrolyte components



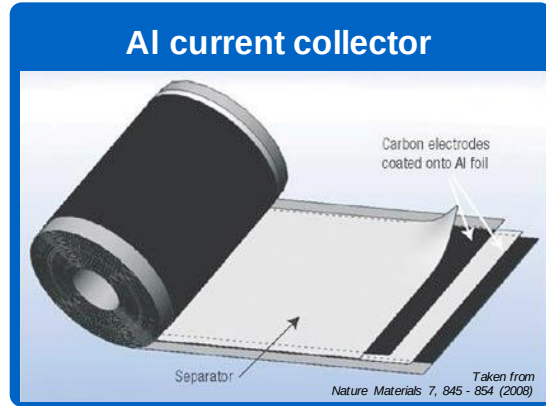
Development of novel electrolyte components



Novel electrolytes for supercapacitor



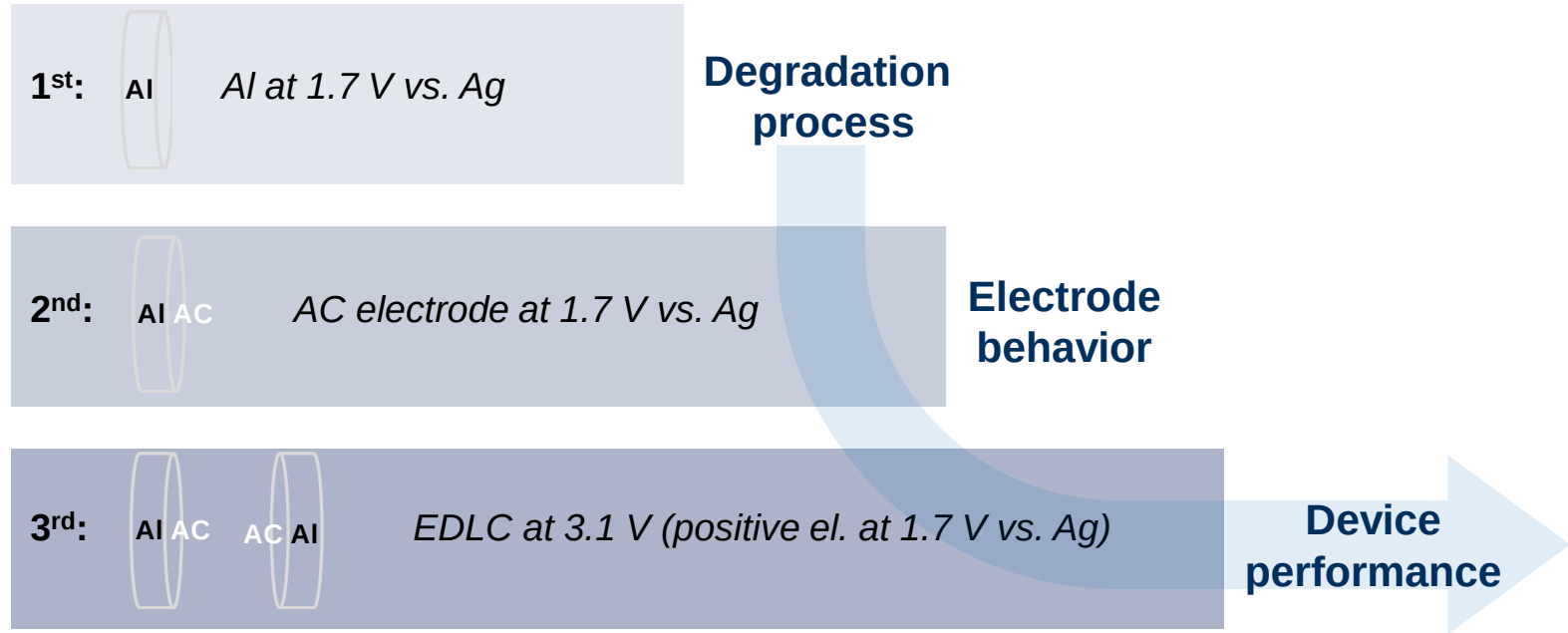
Novel solvents for EDLCs and inactive electrode components



- ❑ The degradation of the current collector has a detrimental effect on the cycling stability of EDLCs
- ❑ The composition of the salt and the ion-solvent interactions strongly affect the anodic dissolution of Al

Impact of novel electrolytes on the anodic dissolution of Al current collectors

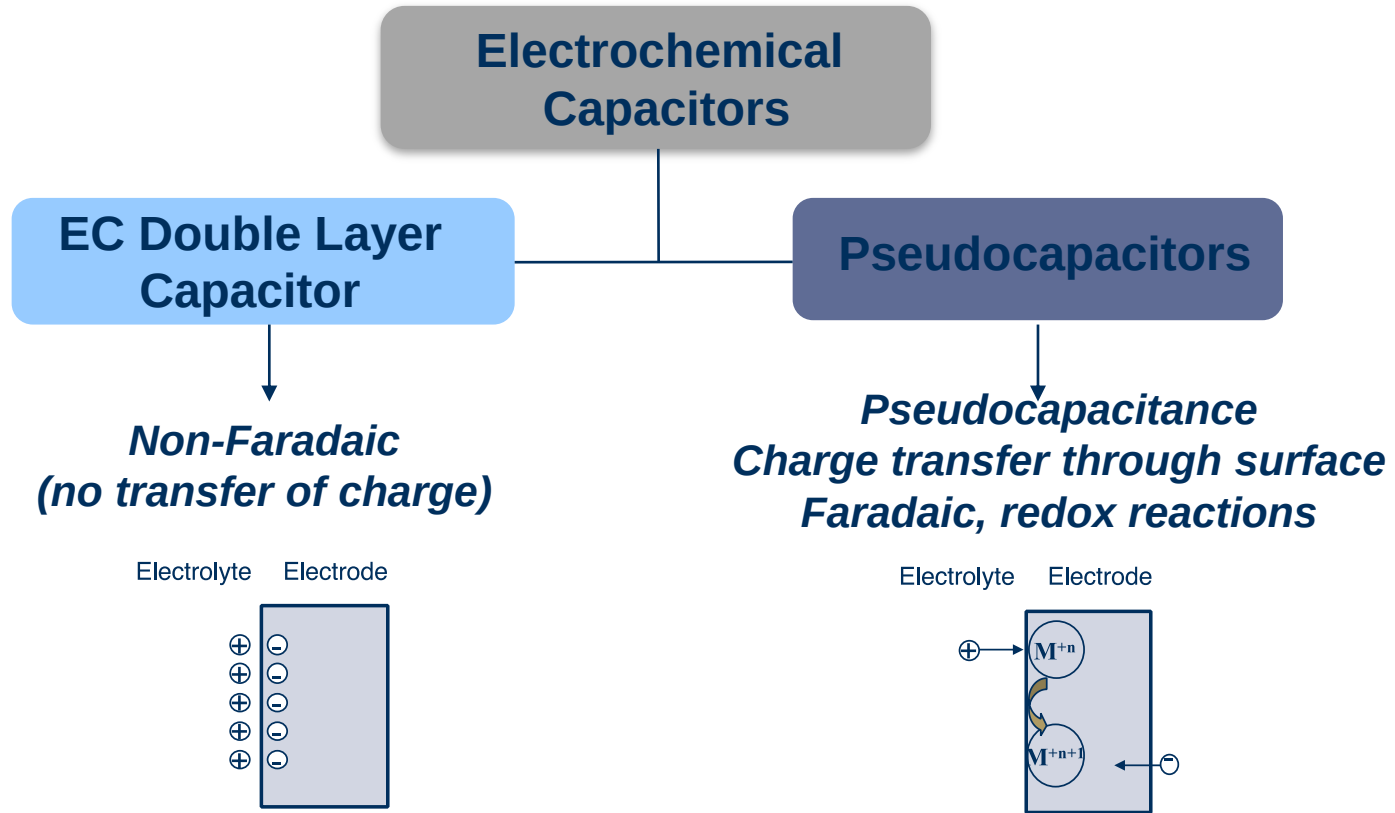
Float tests in selected electrolytes



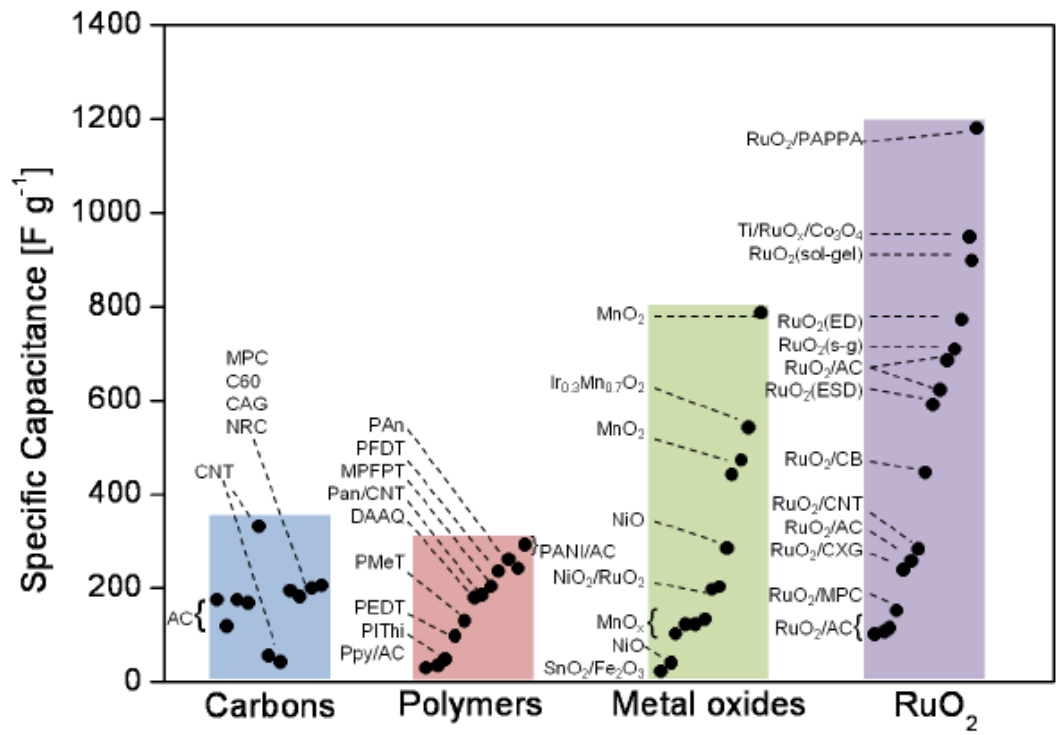
Outline

- High power applications: supercapacitors vs. batteries
- Supercapacitors
 - Components and materials
 - Techniques
- EDLC
 - Carbon
 - Electrolytes
- Pseudocapacitors
- Hybrid devices

Supercapacitors



Active materials in supercapacitors



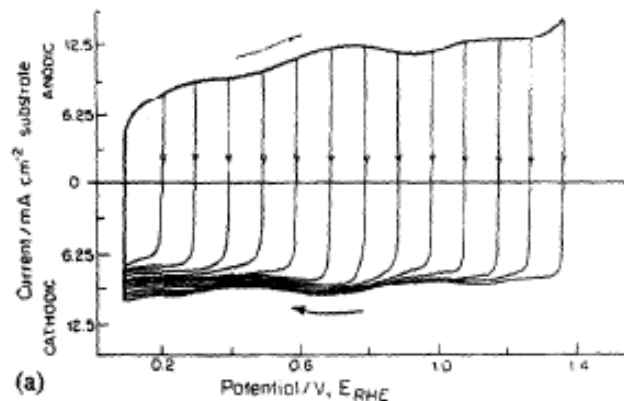
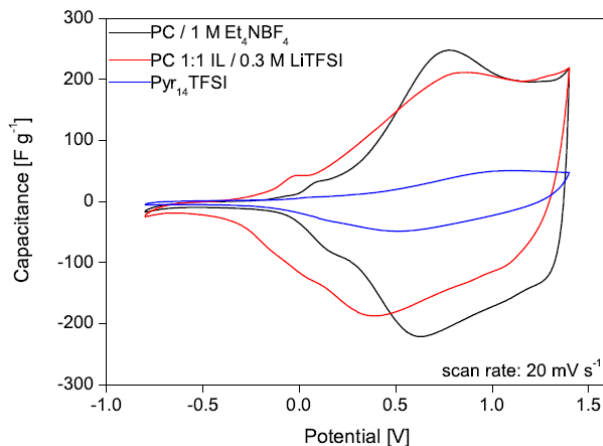
Pseudocapacitive materials

Pseudocapacitance
Charge transfer through surface
Faradaic, redox reactions



- Conducting polymers
- Metal Oxides
- RuO_2

Battery-like electrodes





JES FOCUS ISSUE ON ELECTROCHEMICAL CAPACITORS: FUNDAMENTALS TO APPLICATIONS

To Be or Not To Be Pseudocapacitive?

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^b*Réseau sur le Stockage Electrochimique de l'Energie (RS2E), FR CNRS 3459, France*

^c*Département Chimie, Université du Québec à Montréal, Succursale Centre-Ville, Montréal, Québec H3C 3P8, Canada*

^d*Code 6170, Surface Chemistry Branch, U.S. Naval Research Laboratory, Washington, DC 20375, USA*

In the field of electrochemical capacitors, the term “pseudocapacitance” is used to designate electrode materials (RuO₂, MnO₂) that have the electrochemical signature of a capacitive electrode (such as observed with activated carbon), i.e., exhibiting a linear dependence of the charge stored with the width of the potential window, but where charge storage originates from different reaction mechanisms

Pseudocapacitive materials

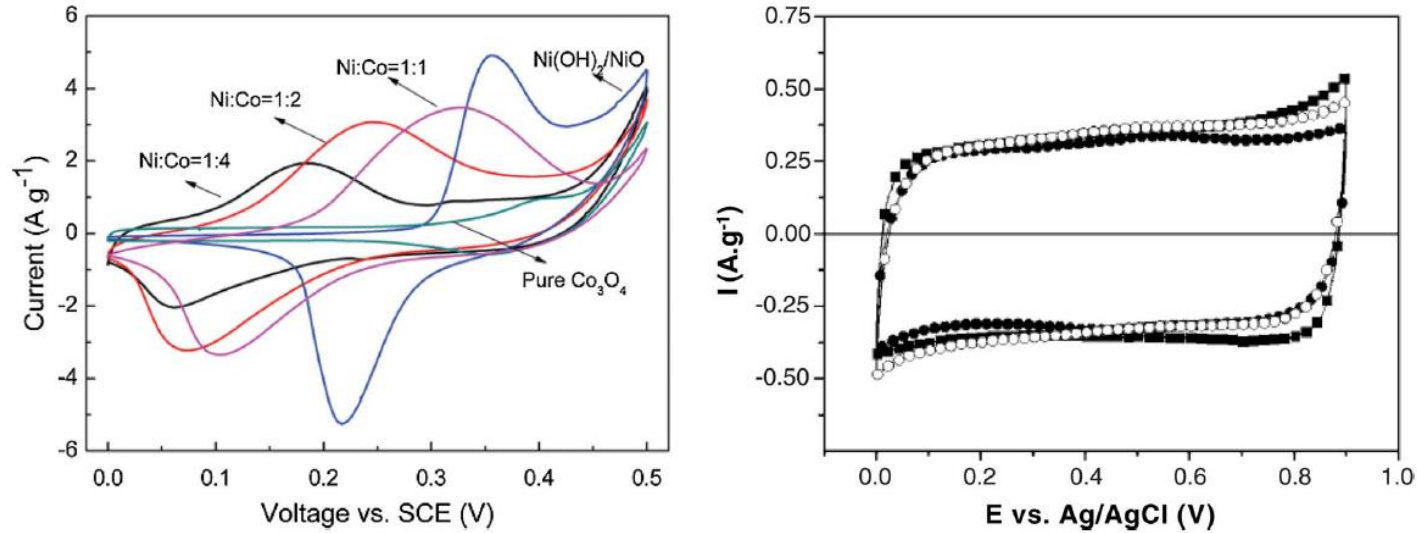
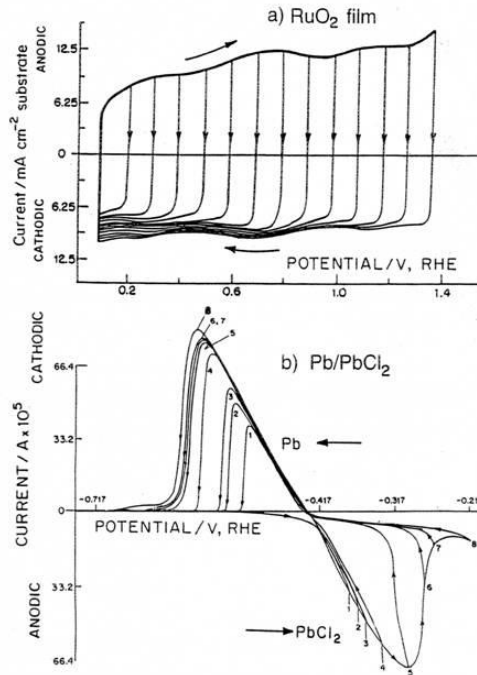


Figure 2. Cyclic voltammograms of Ni(OH)₂ and different cobalt doped nickel oxides⁶ that exhibit pure faradaic behavior, while MnO₂ electrode⁵ shows the rectangular shape CV typical of a pseudocapacitive oxide.

Pseudocapacitive materials



Cyclic voltammogram for RuO₂ in 1 M aqueous H₂SO₄ (298K) showing mirror-image symmetry and responses to successive switching along the scanned potential range

Cyclic voltammogram for Pb-PbCl₂ battery electrode showing typical irreversibility arising with 3-dim materials undergoing chemical phase changes

Pseudocapacitive material: MXenes

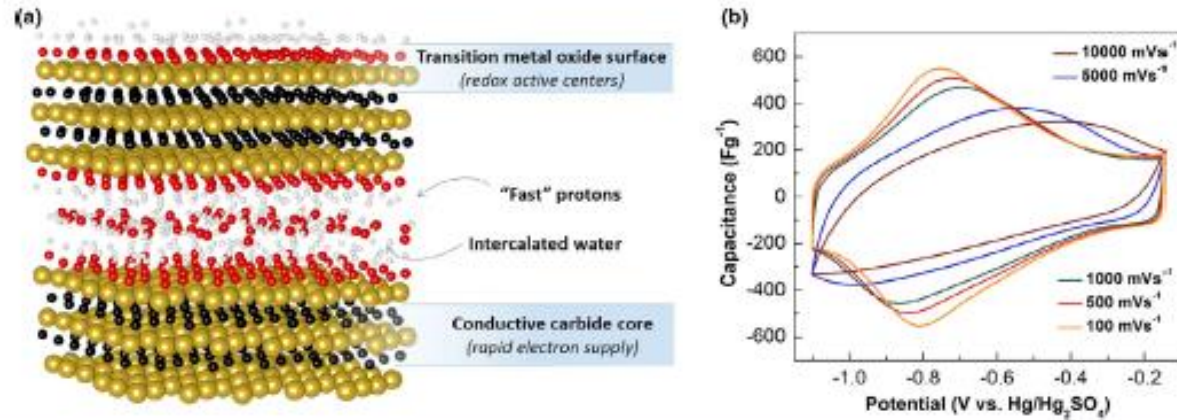


FIGURE 7

(a) Schematic illustration of MXene structure: MXenes possess excellent conductivity owing to a conductive carbide inner layer along with transition metal oxide-like surfaces. Intercalated water molecules enable high accessibility of the redox-active sites on MXene surfaces to ions, i.e., protons; (b) cyclic voltammetry profiles of macro-porous $\text{Ti}_3\text{C}_2\text{T}_x$ film with 0.43 mg cm^{-2} loading and $13\text{-}\mu\text{m}$ thickness collected in $3 \text{ M H}_2\text{SO}_4$ at scan rates from 100 to $10,000 \text{ mV s}^{-1}$. Adapted with permission from Ref. [177].

Pseudocapacitive material: Conducting polymers

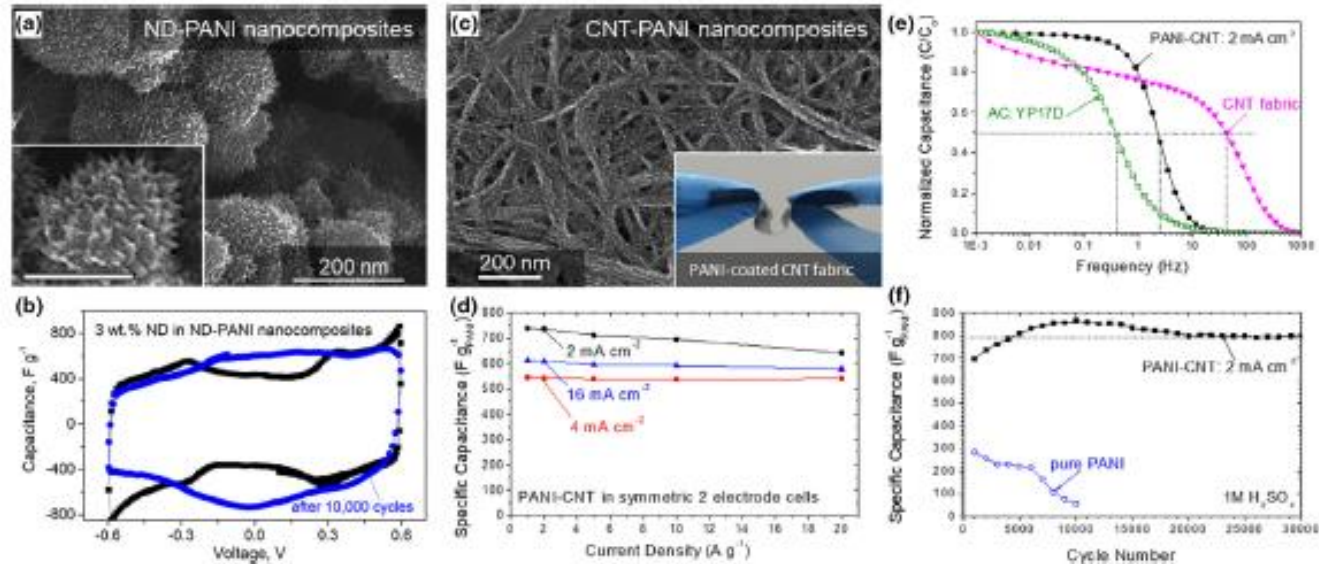


FIGURE 8

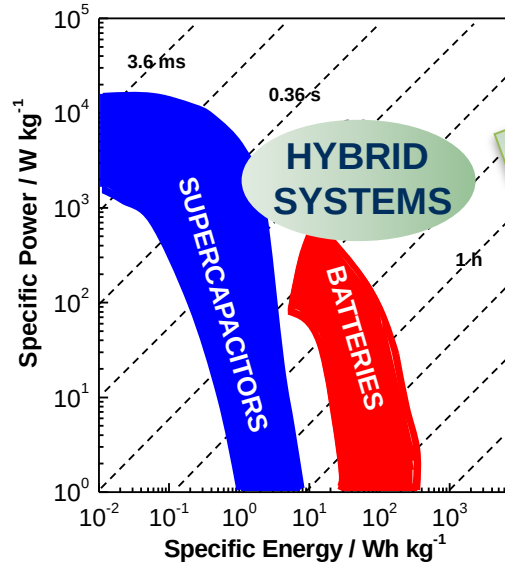
Selected examples of the conductive polymer-carbon composites and their electrochemical behavior, such as (a) SEM micrographs of PANI with 3 wt.% of nanodiamond (ND) soot incorporated into the PANI particles during synthesis; and (b) CV of a symmetric pseudocapacitor with ND-PANI electrodes; (c) SEM of PANI electrodeposited on CNT fabric at the smallest current density of 2 $mA cm^{-2}$; (d) its capacitance retention at high current densities; (e) frequency response in comparison with commonly used AC (YP-17D) and CNT fabric with no PANI; and (f) cycle stability showing over 30,000 stable cycles in H_2SO_4 aqueous electrolyte. Adapted with permission from Refs. [184,190].

Outline

- High power applications: supercapacitors vs. batteries
- Supercapacitors
 - Components and materials
 - Techniques
- EDLC
 - Carbon
 - Electrolytes
- Pseudocapacitors
- Hybrid devices

Hybrid devices

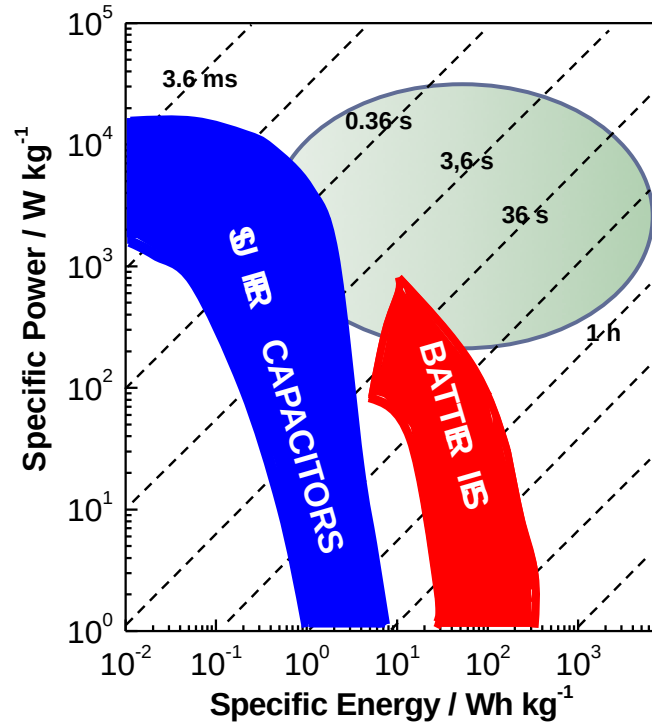
Supercapacitors and batteries
will remain different



High Energy
High Power
Medium-high cycling stability

Hybrid devices

Hybrid devices



New type of high power devices

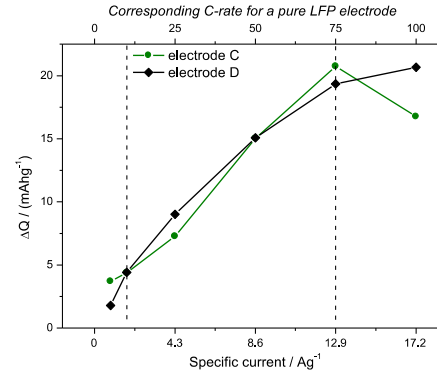
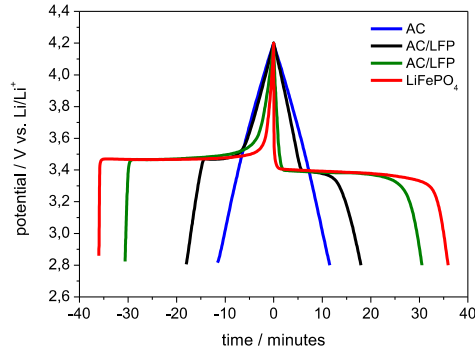
- High energy ($40\text{--}60 \text{ Wh kg}^{-1}$)
(higher than supercaps)
- High power ($5\text{--}8 \text{ kW kg}^{-1}$)
(higher than batteries)
- Medium-high cycling stability
(hundred of thousand cycles)

Hybridization

Active material hybridization

Combination into one electrode of supercapacitors and battery material

Interaction of LiFePO_4 (LFP) and Activated Carbon

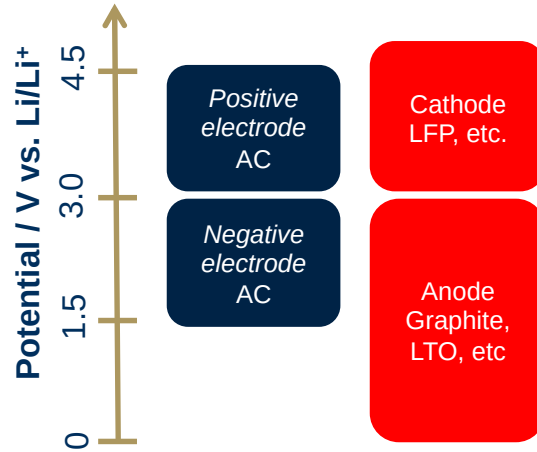


AC improves the performance of LFP at high specific current

Hybridization

Electrode combination

Combination of supercapacitors and battery electrodes into one devices



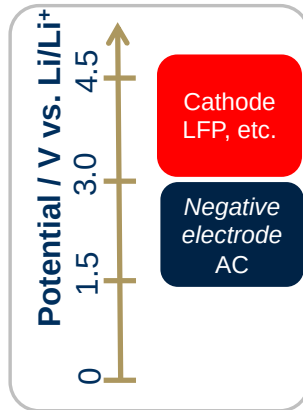
2 processes occur simultaneously during charge-discharge:

- Li insertion-extraction
- Double layer formation and depletion

Hybridization

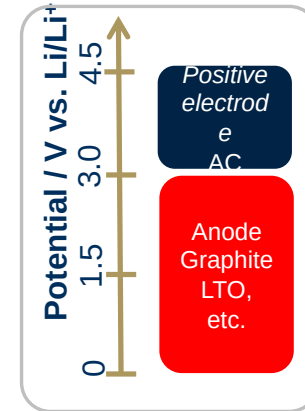
Electrode combination

Type 1



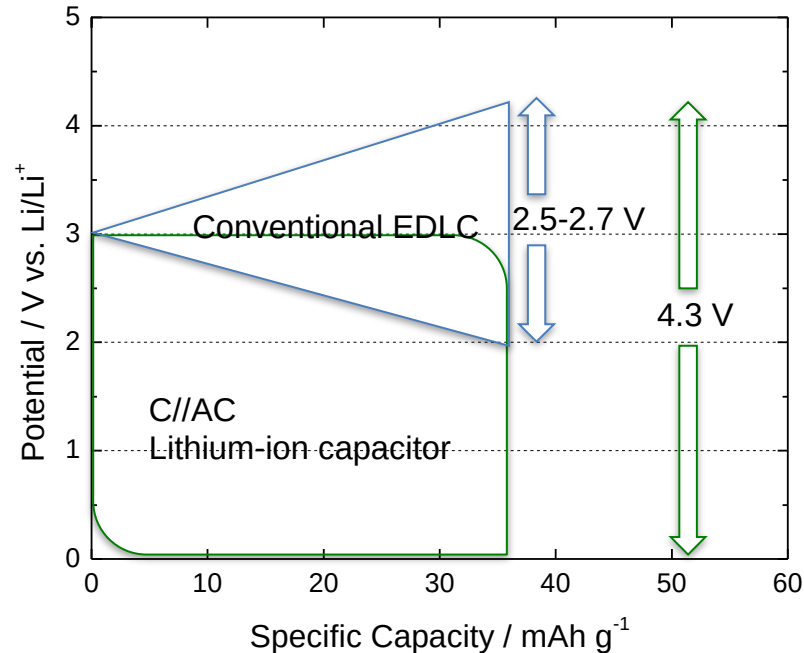
Lower operative voltage
Contain lithium (cathode)

Type 2



Higher operative voltage
Does not contain lithium (pre-lithiation)

Type 2 device: lithium-ion capacitors

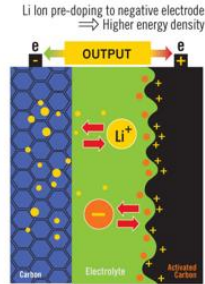
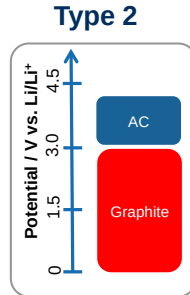


- High voltage (4.3 V)
- High power
- High energy (25 Wh/kg)
- High cycle life (>50.000 cycles)
- Low self discharge

Key technology:
Pre-doping of Li to the carbon anode

Type 2 device: lithium-ion capacitors

Commercially available devices (JSR Micro)



Positive electrode:
Activated carbon

Negative electrode
Graphite

Electrolyte
State-of-the-art LIBs

Graphite displays poor capacity retention during intercalation processes at high currents



Alternative materials for the negative electrode of LIC would be beneficial

Conclusions

- Supercapacitors are important (and commercially available) energy storage devices which are used in a large number of applications
- The energy of supercapacitors should be improved
 - Active materials
 - Electrolytes
 - Inactive components
- We need new ideas and new scientists